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Coordinated Thermodynamically Feasible Production Scheduling and Bidding for Process Industrial Loads Under PV and Price Uncertainties

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HIGHLIGHTS

- Thermodynamic constraints preserve production feasibility in industrial scheduling.
- SOS2-McCormick approximation enables tractable thermodynamic scheduling.
- SC-IGDT coordinates production and bidding under PV and price uncertainties.
- The framework improves economic performance while maintaining production safety.

ABSTRACT

High renewable-energy penetration increases net-load variability and spot-price uncertainty, strengthening the need for flexible industrial demand in electricity markets. However, for process industrial loads (PILs), uncertain photovoltaic (PV) output and electricity prices interact with tightly coupled material and thermodynamic dynamics, linking economic bidding to production feasibility and operational safety. This paper proposes a coordinated production-scheduling and bidding framework for an electrolytic copper load by integrating a thermodynamically enhanced resource-task network (ThE-RTN) with stochastic programming-CVaR-information gap decision theory (SC-IGDT). The ThE-RTN embeds power-temperature-efficiency coupling into plant-wide material-flow scheduling and is reformulated via SOS2-McCormick linearization. SC-IGDT combines scenario-based CVaR for PV uncertainty with IGDT for bounded day-ahead and real-time price uncertainty under settlement and arbitrage-recovery mechanisms. A joint-vertex affine-epigraph reformulation with outer bisection determines the maximum feasible uncertainty radius. Case studies show that a 10% cost tolerance yields a maximum feasible price-uncertainty radius of 3.781%, and the proposed framework reduces spot-exposure risk cost by 17.77% relative to conventional RTN scheduling and bidding. SC-IGDT further provides effective protection against extreme costs and budget violations. Thermodynamic replay confirms production and temperature feasibility, whereas the conventional RTN causes thermal violations and a production shortfall. These results demonstrate effective coordination among economic efficiency, uncertainty protection, and operational safety.

Keywords: Process industrial load (PIL); photovoltaic and price uncertainties; resource-task network (RTN); information gap decision theory (IGDT); production scheduling and bidding.

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1. Introduction

1.1 Background and Motivation

Under the dual-carbon goals and high renewable penetration, renewable variability has increased the need for demand-side flexibility in power grids [1]-[2]. Process industrial loads (PILs) are promising large-scale flexibility resources because of their substantial electricity demand, strong power-regulation capability, material buffers, and thermal inertia [3]-[5]. Market-based demand-response scheduling can mobilize this flexibility by coordinating production schedules with electricity consumption, thereby compensating for renewable variability while supporting secure grid operation [6]. Yet feasible PIL responses cannot be determined from electrical limits alone; they also depend on production continuity, intermediate-material inventories, equipment operating conditions, and the carryover of internal process states across periods [7]-[8]. Economically attractive power adjustments must also respect process dynamics and safe operating limits to be implementable in practice [9]. Accordingly, plant-wide interstage material flows and power-temperature-efficiency dynamics must be represented to translate potential flexibility into operationally feasible and reliable grid support.

1.2 Literature Review and Research Gap

State-Task Network (STN) and Resource-Task Network (RTN) models are widely used in PIL scheduling to represent production tasks, equipment states, material inventories, and inter-stage couplings. Representative applications of process-level scheduling include coordinating steelmaking production with energy and reserve-market bidding [10], and integrating cement production with on-site renewables, storage, and electricity-market participation [11]. Recent work has improved the scalability of RTN-based industrial scheduling through continuous task representations and unified constraint formulations [12]. However, when process behavior is represented mainly by fixed conversion relationships or predefined operating modes, internal-state evolution and its intertemporal effects under power regulation may be insufficiently captured [13]-[14]. Recent studies have incorporated process-state dynamics into STN models, but the resulting nonconvex mixed-integer formulations still require tailored algorithms [15]. Meanwhile, thermodynamics-aware scheduling models have begun to consider temperature-dependent efficiency and safety limits [16]-[18], yet generally focus on specific devices or local processes. Integrating power-temperature-efficiency coupling into a plant-wide scheduling framework with multi-stage material networks while retaining industrial-scale tractability remains an open challenge.

Beyond physical modeling, grid-interactive PILs with distributed photovoltaic (PV) generation also face uncertainties in PV output and spot prices. PV forecast errors can generally be represented by probabilistic scenarios derived from historical samples, whereas spot prices exhibit spikes and nonstationarity, making stable distributions difficult to identify [19]-[20]. In day-ahead-real-time (DA-RT) markets with deviation settlement and arbitrage-recovery rules, these uncertainties may further translate into settlement risk. Existing studies use scenario-based stochastic programming (SP), conditional value-at-risk (CVaR), robust optimization (RO), distributionally robust optimization (DRO), and information-gap decision theory (IGDT) under different information conditions [19],[21]. Applying RO to both uncertainties cannot fully exploit probabilistic PV information, while DRO depends on samples and ambiguity-set construction. By contrast, IGDT evaluates tolerable price deviations under a prescribed cost requirement. Thus, combining SP/CVaR for PV uncertainty with IGDT for price uncertainty better matches their distinct characteristics. A recent study applies CVaR and entropy-weighted IGDT to multi-stage VPP trading in coupled electricity-green markets [22]. However, risk-based market decisions remain insufficiently coordinated with the physical production feasibility of PILs. How to jointly address heterogeneous PV and price uncertainties and complex settlement rules while coordinating production scheduling, market bidding, and settlement risk for the safe, reliable, and economic utilization of PIL flexibility remains an open issue.

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1.3 Contribution and Summary

In summary, the participation of PILs in renewable-integrated grid interactions requires both an accurate characterization of production-constrained executable flexibility and its reliable and economic market utilization under PV and spot-price uncertainties. Using an actual electrolytic copper load (ECL) in Yunnan as the study system, this paper develops a coordinated production-scheduling and bidding optimization method combining a Thermodynamically Enhanced Resource-Task Network (ThE-RTN) with Stochastic Programming-CVaR-Information-Gap Decision Theory (SC-IGDT). The main contributions are as follows:

• A ThE-RTN model is developed by incorporating the power-temperature-efficiency relationship of electrolysis into the RTN, thereby capturing the cross-period effects of power regulation on thermal states, production efficiency, and yield. The resulting nonconvex couplings are handled through a Type 2 (SOS2)-McCormick formulation, yielding an MILP-solvable production scheduling model.

• An SC-IGDT production-scheduling and bidding co-optimization model is formulated for heterogeneous uncertainties of PV and DA-RT prices, while accounting for deviation settlement and arbitrage-recovery mechanisms. Based on the ThE-RTN, the model jointly optimizes DA bids and production schedules and determines the maximum tolerable price deviation under a prescribed spot-exposure cost tolerance.

• To address the SIP induced by the continuous price set and the bilinear coupling between the uncertainty radius and the bidding and actual net-purchase variables, a joint-vertex affine-epigraph reformulation and an outer bisection algorithm are developed. The fixed-radius subproblem is converted into an MILP, yielding the maximum feasible price-uncertainty radius and its associated production schedule and DA bids.

The remainder of this paper is organized as follows. Section 2 establishes the ThE-RTN model and its SOS2-McCormick approximation for the ECL. Section 3 formulates the SC-IGDT co-optimization model for production scheduling and bidding and presents the joint-vertex affine-epigraph reformulation and outer bisection algorithm. Section 4 presents the case studies, and Section 5 concludes the paper.

2. Thermodynamically Enhanced RTN Scheduling Model for PILs

Conventional RTN models cannot fully capture the cross-period effects of power regulation on thermal states, efficiency, and production. Therefore, a ThE-RTN is developed for the ECL by embedding power-temperature-efficiency dynamics into its multi-stage material network, while a hybrid SOS2-McCormick formulation is used to handle the resulting nonconvexities.

The model assumes that: 1) power regulation states change only at interval boundaries, with inter-stage material transfer delays negligible relative to the scheduling interval; and 2) ambient temperature remains constant over the scheduling horizon, while the narrow electrolyte temperature range allows heat-dissipation power to be approximated as constant.

2.1 Process Characteristic Analysis and Power Regulation Cost Modeling of ECL

As a typical PIL, the ECL integrates sequential pyrometallurgical processes and auxiliary units. Raw materials from Material Preparation (MP) undergo progressive smelting-Side-Blowing (SBS), Top-Blowing (TBS), and Anode Furnace Smelting (AFS)-followed by Casting (C) to yield anode copper, which is finally electrolyzed via Electrolytic Refining (ER) into cathode copper. Concurrently, Auxiliary Smelting (AS) and Oxygen Production (OP) supply pyrometallurgical oxygen, while Acid Production (AP) recovers smelting gases.

Power regulation methods for these stages are typically categorized into continuous power regulation and discrete load shedding. Since ECL production scheduling decisions must account for PV output characteristics, the power and regulation characteristics of the equipment under PV scenarios can be formulated as Eqs. (1)-(2) [13]:

$$P^{k,i,t,\omega} = P^{k,i}_{\text{rate}} x^{k,i,t,\omega} s^{k,i,t,\omega} = P^{k,i}_{\text{rate}} y^{k,i,t,\omega} \quad \forall k,i,t,\omega \tag{1}$$

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$$\begin{cases} \underline{x}^{k,i} \le x^{k,i,t,\omega} \le \overline{x}^{k,i}, s^{k,i,t,\omega} \in \{0,1\} & k \in K_{\mathrm{c}} \\ \underline{x}^{k,i} \le x^{k,i,t,\omega} \le \overline{x}^{k,i}, s^{k,i,t,\omega} = 1 & k \in K_{\mathrm{us}} \\ x^{k,i,t,\omega} = s^{k,i,t,\omega} = 1 & k \in K_{\mathrm{uc}} \end{cases} \tag{2}$$

where $P^{k,i,t,\omega}$, $P^{k,i}_{\mathrm{rate}}$, $x^{k,i,t,\omega}$, $s^{k,i,t,\omega}$, $y^{k,i,t,\omega}$ denote the equipment power, rated power, continuous power regulation state, commitment status, and overall power regulation state, respectively. k, i, t, ω represent the indices for the process stage, equipment, 15-minute time interval, and PV output scenario, respectively. $\underline{x}^{k,i}, \overline{x}^{k,i}$ denotes the power regulation boundaries. Additionally, K_{c}={MP,ER} and K_{us}={OP,AS,AP} denote the sets of controllable and non-sheddable stages, respectively, while the remaining uncontrollable stages belong to the set K_{uc}. Eq. (1) indicates that power consumption depends on distinct power regulation methods, while Eq. (2) captures the power regulation characteristics of different process stages. Since the discrete variables in Eq. (1) render the model non-convex, a new variable $y^{k,i,t,\omega}$ is introduced to represent the overall power regulation state of the equipment, which is then linearized via the tight Big-M method, as formulated in Eq. (3):

$$\begin{cases} y^{k,i,t,\omega} \le \overline{x}^{k,i} s^{k,i,t,\omega} \\ y^{k,i,t,\omega} \le x^{k,i,t,\omega} - \underline{x}^{k,i}(1 - s^{k,i,t,\omega}) \\ y^{k,i,t,\omega} \ge x^{k,i,t,\omega} - \overline{x}^{k,i}(1 - s^{k,i,t,\omega}) \\ y^{k,i,t,\omega} \ge \underline{x}^{k,i} s^{k,i,t,\omega} \end{cases} \quad \forall k,i,t,\omega \tag{3}$$

The scheduling cost induced by power regulation is the core factor constraining the optimal scheduling strategy of the ECL. It comprises continuous regulation capacity costs and discrete start-up/shut-down costs, formulated as Eqs. (4)-(9):

$$F^{h,\omega}_{\mathrm{cpr}} = \sum_{t \in T_h} \sum_{k \in K} \sum_{i \in I_k} P^{k,i}_{\mathrm{rate}} (\Delta y^{k,i,t,\omega}_{\mathrm{up}} \eta^{k,i}_{\mathrm{up}} + \Delta y^{k,i,t,\omega}_{\mathrm{down}} \eta^{k,i}_{\mathrm{down}}) \quad \forall h,\omega \tag{4}$$

$$F^{h,\omega}_{\mathrm{ud}} = \sum_{t \in T_h} \sum_{k \in K} \sum_{i \in I_k} (u^{k,i,t,\omega} \eta^{k,i}_{\mathrm{su}} + d^{k,i,t,\omega} \eta^{k,i}_{\mathrm{sd}}) \quad \forall h,\omega \tag{5}$$

$$y^{k,i,t,\omega} - 1 = \Delta y^{k,i,t,\omega}_{\mathrm{up}} - \Delta y^{k,i,t,\omega}_{\mathrm{down}} \quad \forall k,i,t,\omega \tag{6}$$

$$\Delta y^{k,i,t,\omega}_{\mathrm{up}}, \Delta y^{k,i,t,\omega}_{\mathrm{down}} \ge 0 \quad \forall k,i,t,\omega \tag{7}$$

$$u^{k,i,t,\omega} - d^{k,i,t,\omega} = s^{k,i,t,\omega} - s^{k,i,t-1,\omega} \quad \forall k,i,t,\omega \tag{8}$$

$$u^{k,i,t,\omega} + d^{k,i,t,\omega} \le 1 \quad \forall k,i,t,\omega \tag{9}$$

where $F^{h,\omega}_{\mathrm{cpr}}$, $F^{h,\omega}_{\mathrm{ud}}$ denote the continuous power regulation cost and discrete start-up/shut-down cost, respectively. h, T_h, K, I_k represent the 1-hour time interval index, the set of time periods t within hour h, the set of process stages, and the set of units i within stage k. $\Delta y^{k,i,t,\omega}_{\mathrm{up}}$, $\Delta y^{k,i,t,\omega}_{\mathrm{down}}$ denote the continuous upward and downward regulation amounts, with $\eta^{k,i}_{\mathrm{up}}$, $\eta^{k,i}_{\mathrm{down}}$ being their unit costs. $\eta^{k,i}_{\mathrm{su}}$, $\eta^{k,i}_{\mathrm{sd}}$ represent the equipment start-up and shut-down costs, while $u^{k,i,t,\omega}$, $d^{k,i,t,\omega}$ indicate the corresponding start-up and shut-down states. Eqs. (6)-(7) calculate the specific upward and downward regulation capacities within time period t. Eq. (8) determines the start-up and shut-down states, while Eq. (9) prevents simultaneous start-up and shut-down actions.

While Eqs. (1)-(9) quantify ECL regulation and costs, defining actual operational bounds requires inter-stage material and intra-stage thermodynamic couplings. Accordingly, the next section embeds these into the conventional RTN to formulate the ThE-RTN model.

2.2 Formulation of the Enhanced RTN Scheduling Model Incorporating Thermodynamic Coupling Mechanisms

Material interactions in the ECL are constrained by their physical states. For instance, molten materials in

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pyrometallurgical stages cannot be exchanged across parallel production lines. Accordingly, the interaction dynamics for different materials are formulated in Eqs. (10)-(12), and the conventional RTN model for the ECL is illustrated in Fig. 1.

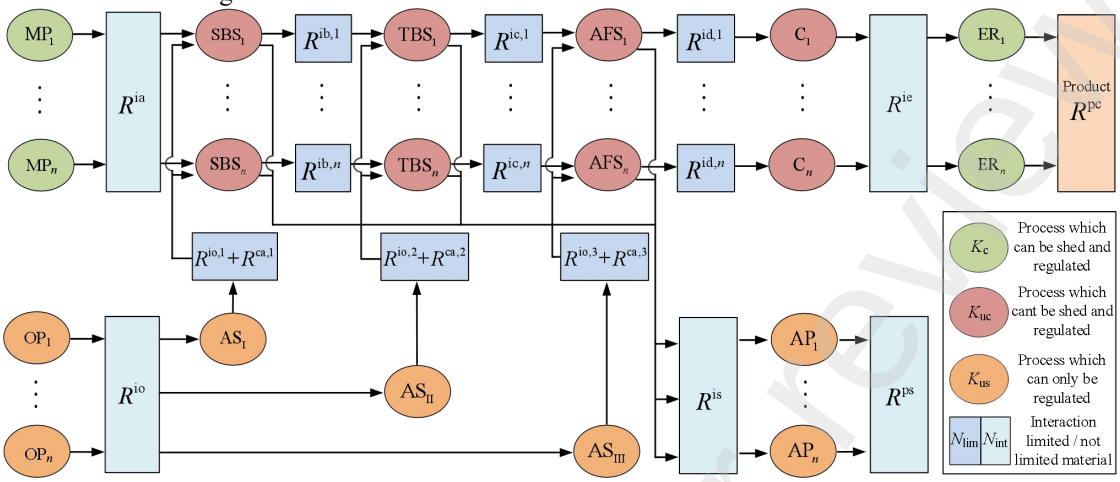


Fig. 1. Conventional RTN model for the ECL. R^{ia}, R^{ib}, R^{ic}, R^{id}, R^{ie}, R^{io}, R^{ca}, R^{is}, R^{ps}, and R^{pc} represent raw materials, blister copper, matte, anode copper, solid anode plates, oxygen, compressed air, sulfur-containing gas, sulfuric acid, and product cathode copper, respectively. The restricted and unconstrained material sets are denoted by N_{lim}={ib,ic,id,ca} and N_{int}={ia,ie,io,is,ps}, respectively.

$$R^{n,t,\omega} = R^{n,t-1,\omega} + \sum_{k\in K}\sum_{i\in I_k} v^{n,k,i} y^{k,i,t,\omega} \quad \forall n \in N_{\text{int}}, \omega \tag{10}$$

$$R^{n,t,\omega} = \sum_{i\in I_k} (R^{n,i,t-1,\omega} + \sum_{k\in K} v^{n,k,i} y^{k,i,t,\omega}) \ \forall n \in N_{\text{lim}}, \omega \tag{11}$$

$$\underline{R}^{n} \le R^{n,t,\omega} \le \overline{R}^{n} \quad \forall n,t,\omega \tag{12}$$

where $R^{n,t,\omega}$, $v^{n,k,i}$ denote the material inventory and its production or consumption rate. n, N_{int}, N_{lim} represent the material index, the set of materials unconstrained by production lines, and the set of materials restricted to specific lines. $\underline{R}^{n}, \overline{R}^{n}$ denotes the material storage boundaries.

For the ER stage characterized by thermal inertia, the production efficiency of R^{pc} depends on electrolytic power, electrolyte concentration, and temperature [18]. Since continuous replenishment maintains a stable concentration in practice, this paper focuses on the complex "power-temperature-efficiency" coupling. As indicated in [18], electrolytic efficiency approximates a linear function of temperature within a narrow operational band. Consequently, the electrolyte temperature, its boundaries, and temperature-efficiency coupling are formulated as Eqs. (13)-(15):

$$T^{\text{ER},i,t,\omega} = T^{\text{ER},i,t-1,\omega} + \frac{\Delta t}{c^{\text{ER},i} m^{\text{ER},i}} (P^{\text{ER},i,t,\omega} x_{\text{heat}} - Q^{\text{ER},i}) \quad \forall t,\omega \tag{13}$$

$$\underline{T}^{\text{ER},i} \le T^{\text{ER},i,t,\omega} \le \overline{T}^{\text{ER},i} \quad \forall t,\omega \tag{14}$$

$$\eta^{\text{ER},i,t,\omega} = \eta^{\text{ER},i}_{\text{rate}} - e^{\text{ER},i} (T^{\text{ER},i}_{\text{rate}} - T^{\text{ER},i,t,\omega}) \quad \forall t,\omega \tag{15}$$

where $T^{\text{ER},i,t,\omega}$, $T^{\text{ER},i}_{\text{rate}}$, $Q^{\text{ER},i}$, x_{heat} denote the electrolyte temperature, rated temperature, heat dissipation power, and heat conversion efficiency, respectively. Δt represents the 15-minute time interval. $c^{\text{ER},i}$,$m^{\text{ER},i}$ are the specific heat capacity and mass of the electrolyte, with $\underline{T}^{\text{ER},i}$,$\overline{T}^{\text{ER},i}$ indicating its temperature boundaries. Furthermore, $\eta^{\text{ER},i,t,\omega}$,$\eta^{\text{ER},i}_{\text{rate}}$, $e^{\text{ER},i}$ denote the electrolytic efficiency, rated electrolytic efficiency, and temperature-efficiency coefficient, respectively.

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Thus, the equipment- and temperature-dependent R^{pc} yield and production constraints are formulated as Eqs. (16)-(17):

$$R^{\mathrm{pc},t,\omega} = R^{\mathrm{pc},t-1,\omega} + \sum_{i\in I_{\mathrm{ER}}} v^{\mathrm{pc},\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} \eta^{\mathrm{ER},i,t,\omega} \quad \forall t,\omega \tag{16}$$

$$R^{\mathrm{pc},T,\omega} - R^{\mathrm{pc},0,\omega} \geq R_{\mathrm{aim}} \quad \forall \omega \tag{17}$$

where $R^{\mathrm{pc},T,\omega}$, $R^{\mathrm{pc},0,\omega}$ denote the final and initial R^{pc} inventory of the production cycle, while R_{aim} represents the target yield.

To embed the temperature-dependent production efficiency into the RTN, an auxiliary variable $z^{\mathrm{ER},i,t,\omega}$ is introduced to represent the product $y^{\mathrm{ER},i,t,\omega}\eta^{\mathrm{ER},i,t,\omega}$. Combining the power-temperature dynamics in Eq. (13) with the efficiency relation in Eq. (15) gives Eq. (18):

$$\begin{aligned} z^{\mathrm{ER},i,t,\omega} &= y^{\mathrm{ER},i,t,\omega}\eta^{\mathrm{ER},i,t,\omega} \\ &= \frac{P_{\mathrm{rate}}^{\mathrm{ER},i} e^{\mathrm{ER},i} x_{\mathrm{heat}} \Delta t}{c^{\mathrm{ER},i} m^{\mathrm{ER},i}} (y^{\mathrm{ER},i,t,\omega})^2 + e^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} T^{\mathrm{ER},i,t-1,\omega} + (\eta_{\mathrm{rate}}^{\mathrm{ER},i} - e^{\mathrm{ER},i} T_{\mathrm{rate}}^{\mathrm{ER},i} - \frac{Q^{\mathrm{ER},i,t} e^{\mathrm{ER},i} \Delta t}{c^{\mathrm{ER},i} m^{\mathrm{ER},i}}) y^{\mathrm{ER},i,t,\omega} \\ &= A^{\mathrm{ER},i} (y^{\mathrm{ER},i,t,\omega})^2 + E^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} T^{\mathrm{ER},i,t-1,\omega} + F^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} \end{aligned} \tag{18}$$

where $A^{\mathrm{ER},i}$, $E^{\mathrm{ER},i}$, $F^{\mathrm{ER},i}$ are constant coefficients determined by the production and thermodynamic parameters. The resulting coupling contains a quadratic term and an intertemporal bilinear term, which turns the RTN formulation into a nonconvex MINLP and may hinder reliable solution. A targeted linearization framework is therefore developed to obtain a tractable mixed-integer linear formulation.

2.3 Hybrid SOS2-McCormick Linearization Framework for ThE-RTN Non-convex Couplings

As previously analyzed, R^{pc} efficiency involves quadratic power terms and time-varying bilinearities. Accordingly, a hybrid linearization framework is proposed to ensure both computational tractability and physical accuracy. Specifically, the SOS2 is employed for high-precision linearization of the quadratic terms. For the bilinear terms, McCormick envelopes are utilized for accurate convex relaxation, leveraging the independent boundaries of temperature and power across adjacent intervals. The framework is formulated as Eq. (19).

In the first part, the feasible region of power regulation states is given by Eq. (20). The SOS2 method approximates a nonlinear point as a convex combination of adjacent vertices. Specifically, the non-zero power interval is discretized into N_{p} ordered breakpoints in set $\{\mathbb{Y}_1, \mathbb{Y}_2, ...\mathbb{Y}_{N_{\mathrm{p}}}\}$. Evaluating function $f(y^{\mathrm{ER},i})=A^{\mathrm{ER},i}y^{\mathrm{ER},i2}+F^{\mathrm{ER},i}y^{\mathrm{ER},i}$ at these points yields set $\{\mathbb{Z}_1, \mathbb{Z}_2, ...\mathbb{Z}_{N_{\mathrm{p}}}\}$, where $\mathbb{Z}_j = f(\mathbb{Y}_j)$. By introducing non-negative weights and strictly restricting at most two adjacent weights to be non-zero, the quadratic terms are accurately approximated. The reconstructed constraints are formulated as Eqs. (21)-(24):

$$z^{\mathrm{ER},i,t,\omega} = \underbrace{[A^{\mathrm{ER},i}(y^{\mathrm{ER},i,t,\omega})^2 + F^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega}]}_{\text{Part 1: SOS2}} + \underbrace{E^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} T^{\mathrm{ER},i,t-1,\omega}}_{\text{Part 2: M Cormick}} \tag{19}$$

$$y^{\mathrm{ER},i,t,\omega} \in \{0\} \cup [\underline{y}^{\mathrm{ER},i}, \overline{y}^{\mathrm{ER},i}] \quad \forall t,\omega \tag{20}$$

$$0 \leq \lambda^{i,j,t,\omega} \leq s^{\mathrm{ER},i,t,\omega} \quad \forall j \in \{1,...,N_{\mathrm{p}}\}, t, \omega \tag{21}$$

$$y^{\mathrm{ER},i,t,\omega} = \sum_{j=1}^{N_{\mathrm{p}}} \lambda^{i,j,t,\omega} \cdot \mathbb{Y}^{i,j} \quad \forall t,\omega \tag{22}$$

$$Z_{\mathrm{fix}}^{\mathrm{ER},i,t,\omega} = \sum_{j=1}^{N_{\mathrm{p}}} \lambda^{i,j,t,\omega} \cdot \mathbb{Z}^{i,j,\omega} \quad \forall t,\omega \tag{23}$$

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$$\sum_{j=1}^{N_{\mathrm{p}}} \lambda^{i,j,t,\omega} = s^{\mathrm{ER},i,t,\omega} \quad \{\lambda^{i,1,t,\omega},\ldots,\lambda^{i,N_{\mathrm{p}},t,\omega}\} \in \mathrm{SOS2} \tag{24}$$

where j denotes the ordered breakpoint index, while $\lambda^{i,j,t,\omega}$, $Z_{\mathrm{fix}}^{\mathrm{ER},i,t,\omega}$ represent the weight coefficients and the combined approximation. Eq. (21) bounds the interpolation weights. Eqs. (22) and (23) reconstruct the power-regulation state and the function value, respectively. Eq. (24) links the weight sum to the commitment status and imposes the SOS2 condition, allowing at most two adjacent weights to be non-zero and forcing all weights to zero when the unit is off.

Utilizing McCormick envelopes for the bounded independent bilinearities, the relaxation constraints and the resulting linearized efficiency are given by Eqs. (25)-(26):

$$\begin{cases} W^{\mathrm{ER},i,t,\omega} \geq \underline{T}^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} \\ W^{\mathrm{ER},i,t,\omega} \geq \overline{T}^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} + \overline{y}^{\mathrm{ER},i} T^{\mathrm{ER},i,t-1,\omega} - \overline{T}^{\mathrm{ER},i} \overline{y}^{\mathrm{ER},i} \\ W^{\mathrm{ER},i,t,\omega} \leq \underline{T}^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} + \overline{y}^{\mathrm{ER},i} T^{\mathrm{ER},i,t-1,\omega} - \underline{T}^{\mathrm{ER},i} \overline{y}^{\mathrm{ER},i} \\ W^{\mathrm{ER},i,t,\omega} \leq \overline{T}^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} \end{cases} \quad \forall t,\omega \tag{25}$$

$$z^{\mathrm{ER},i,t,\omega} = Z_{\mathrm{fix}}^{\mathrm{ER},i,t,\omega} + E^{\mathrm{ER},i} \cdot W^{\mathrm{ER},i,t,\omega} \quad \forall t,\omega \tag{26}$$

where $W^{\mathrm{ER},i,t,\omega}$ denotes the auxiliary variable introduced by the McCormick relaxation. Eqs. (20)-(26) form the SOS2-McCormick framework for the 'power-temperature-efficiency' coupling, successfully linearizing R^{pc} efficiency.

The overall error bound $e_{\mathrm{S\text{-}C}}$ of this framework comprises the SOS2 interpolation error e_{S} and the McCormick relaxation error e_{C}, with its theoretical upper limit given by Eq. (27):

$$e_{\mathrm{S\text{-}C}}^{i} \leq \underbrace{\frac{1}{8} h^2 \max |f''(y^{\mathrm{ER},i})|}_{\text{Part 1: } e_{\mathrm{S}}} + \underbrace{\frac{|E^{\mathrm{ER},i}|}{4} \Delta y^{\mathrm{ER},i} \Delta T^{\mathrm{ER},i}}_{\text{Part 2: } e_{\mathrm{C}}} \tag{27}$$

where h denotes the discretization step size, while $\Delta y^{\mathrm{ER},i}$, $\Delta T^{\mathrm{ER},i}$ represent the interval lengths of power bound and temperature. Since the first part is quadratic, the SOS2 error depends on h with an $O(h^2)$ convergence order. Given the narrow operational bounds of both power and temperature, the upper error bound of this framework is tightly constrained.

Collecting the operating, material-state, thermodynamic state, and auxiliary variables introduced above, the ThE-RTN physical scheduling decision vector $\boldsymbol{X}_{\mathrm{phy}}$ is defined by Eq. (28), where the bold symbols collect the corresponding decision variables over all applicable indices.

$$\boldsymbol{X}_{\mathrm{phy}} = \mathrm{col}\{\mathbf{x}, \mathbf{y}, \mathbf{s}, \Delta\mathbf{y}, \mathbf{u}, \mathbf{d}, \mathbf{R}, \mathbf{T}, \boldsymbol{\eta}, \mathbf{z}, \boldsymbol{\lambda}^{\mathrm{SOS2}}, \mathbf{Z}, \mathbf{W}\} \tag{28}$$

Accordingly, the proposed ThE-RTN combines plant-wide inter-stage material constraints with the ER-specific power-temperature-efficiency dynamics. The former preserves material-network feasibility, whereas the latter captures the cross-period effects of power regulation on temperature and production yield. The hybrid SOS2-McCormick formulation provides an MILP-representable approximation whose error is explicitly bounded by Eq. (27). The variable vector $\boldsymbol{X}_{\mathrm{phy}}$, together with Eqs. (1)-(26), defines the physical scheduling feasible domain used by the subsequent SC-IGDT model. The conventional RTN is extensible across process industries through reconfiguration of task-resource relationships. Since the proposed ThE-RTN retains this network structure and augments it with process-specific thermodynamic couplings, it preserves the structural extensibility of RTN; application to another thermal-inertia PIL requires reconfiguration of the RTN topology and identification of the corresponding thermodynamic equations and parameters. The resulting ThE-RTN framework is illustrated in Fig. 2.

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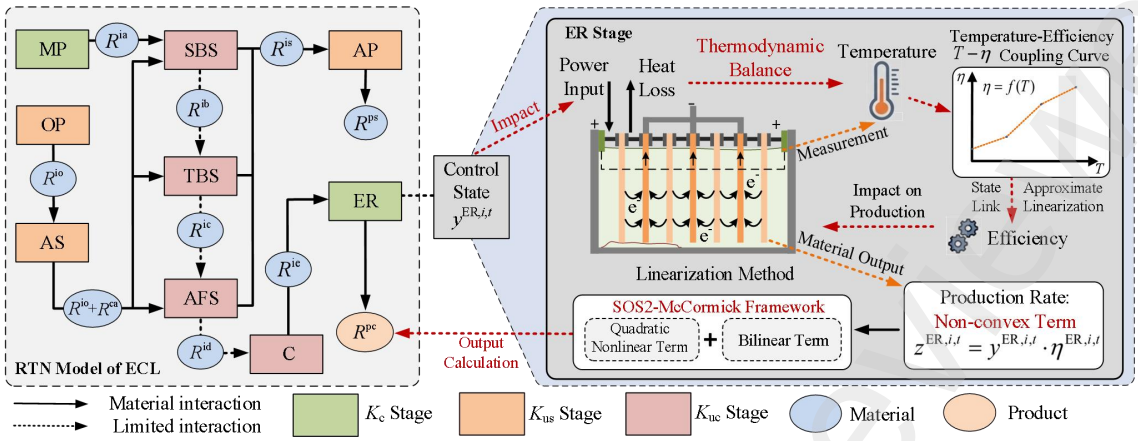


Fig. 2. ThE-RTN model and algorithmic topology for PILs.

3. SC-IGDT Co-optimization of Production Scheduling and Bidding under PV-Price Uncertainties

Building on the ThE-RTN, this section develops an SC-IGDT model for coordinated production scheduling and bidding for ECL participation in the DA-RT spot market under PV and price uncertainties, considering deviation settlement and arbitrage-recovery rules. Scenario-based SP-CVaR and IGDT are used to address PV and price uncertainties, respectively. A joint-vertex affine-epigraph reformulation and outer bisection are employed to handle the SIP and bilinear terms, respectively.

The model assumes that: 1) the ECL is a price taker and does not affect market-clearing prices; 2) PV uncertainty is represented by historical scenarios, whereas DA-RT price uncertainty is described by an IGDT uncertainty set; and 3) the ECL submits volume-only DA bids, while existing mid-to-long-term (MLT) contract quantities and prices are fixed and used only for deviation settlement.

3.1 DA-RT Settlement and PV-price Uncertainty Modeling

Spot settlement for PILs spans three temporal stages: MLT, DA, and RT. Specifically, MLT contracts lock in baseline volumes and prices. DA bidding schedules next-day consumption, with deviations from MLT volumes settled at DA prices. Finally, RT operations execute the schedules, clearing actual consumption deviations from DA bids at RT prices. The multi-stage costs are formulated as Eqs. (29)-(33):

$$C^{h}_{\mathrm{MLT}} = C^{h}_{\mathrm{BAS}} \alpha_{\mathrm{MLT}} \quad \forall h \tag{29}$$

$$F^{h}_{\mathrm{MLT}} = C^{h}_{\mathrm{MLT}} \lambda^{h}_{\mathrm{MLT}} \quad \forall h \tag{30}$$

$$F^{h}_{\mathrm{DA,Dev}} = (C^{h}_{\mathrm{BID}} - C^{h}_{\mathrm{MLT}}) \lambda^{h}_{\mathrm{DA}} \quad \forall h \tag{31}$$

$$F^{h,\omega}_{\mathrm{RT,Dev}} = (C^{h,\omega}_{\mathrm{ACT}} - C^{h}_{\mathrm{BID}}) \lambda^{h}_{\mathrm{RT}} \quad \forall h, \omega \tag{32}$$

$$C^{h,\omega}_{\mathrm{ACT}} = \sum_{t \in I_h} \Big(\sum_{k \in K} \sum_{i \in I_k} P^{k,i,t,\omega} - P^{t,\omega}_{\mathrm{PV}} \Big) \Delta t \quad \forall h, \omega \tag{33}$$

where C^{h}_{MLT}, C^{h}_{BID}, $C^{h,\omega}_{\mathrm{ACT}}$ denote the volumes for MLT contracts, DA bids, and RT actual consumption, respectively; C^{h}_{BAS}, α_{MLT} represent the rated power and MLT volume ratio. $\lambda^{h}_{\mathrm{MLT}}$, $\lambda^{h}_{\mathrm{DA}}$, $\lambda^{h}_{\mathrm{RT}}$ denote the prices for the MLT, DA, and RT stages. F^{h}_{MLT}, $F^{h}_{\mathrm{DA,Dev}}$, $F^{h,\omega}_{\mathrm{RT,Dev}}$ are the costs for the MLT and DA/RT deviation settlements, while $P^{t,\omega}_{\mathrm{PV}}$ is the PV output. Eqs. (29)-(30) formulate the MLT baseline lock-in. Eqs. (31)-(32) define the DA and RT deviation settlements, and Eq. (33) reflects the PV-induced uncertainty in actual RT consumption.

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Furthermore, spot markets impose deviation limits and recover settlement gains arising from excessive deviations [23]. PILs face asymmetric arbitrage-recovery penalties if RT consumption exceeds the penalty-free deadband of DA bids. Specifically, this encompasses over-bidding (over-reporting DA volumes during high RT prices) and under-bidding (under-reporting DA volumes during low RT prices) arbitrages. Accordingly, the penalty formulations are given by Eqs. (34)-(36):

$$\Delta\lambda_{\mathrm{ob}}^{h} = \max(0, \lambda_{\mathrm{RT}}^{h} - \lambda_{\mathrm{DA}}^{h}) \quad \forall h, \omega \tag{34}$$

$$\Delta\lambda_{\mathrm{ub}}^{h} = \max(0, \lambda_{\mathrm{DA}}^{h} - \lambda_{\mathrm{RT}}^{h}) \quad \forall h, \omega \tag{35}$$

$$\begin{aligned} F_{\mathrm{PEN}}^{h,\omega} = \mu_{\mathrm{PEN}} [&\max(0, C_{\mathrm{BID}}^{h} - (1+V_{\mathrm{PEN}}) C_{\mathrm{ACT}}^{h,\omega}) \cdot \Delta\lambda_{\mathrm{ob}}^{h} \\ &+ \max(0, (1-V_{\mathrm{PEN}}) C_{\mathrm{ACT}}^{h,\omega} - C_{\mathrm{BID}}^{h}) \cdot \Delta\lambda_{\mathrm{ub}}^{h}] \end{aligned} \quad \forall h, \omega \tag{36}$$

where $\Delta\lambda_{\mathrm{ob}}^{h}$, $\Delta\lambda_{\mathrm{ub}}^{h}$ denotes the price spreads for over/under-bidding, $F_{\mathrm{PEN}}^{h,\omega}$ is the arbitrage penalty, while μ_{PEN}, V_{PEN} represent the recovery and penalty-free ratios, respectively.

To characterize the PV and price uncertainties involved in Eqs. (33)-(36), scenario-based stochastic modeling and IGDT are adopted for PV output and DA-RT prices, respectively. For PV modeling, Monte Carlo simulation first generates raw trajectories from historical forecast errors around a data-driven baseline forecast. K-medoids clustering is then employed for scenario reduction, with actual sampled trajectories retained as medoids to preserve temporal dependence and physical consistency. The resulting typical PV scenarios and their probabilities are formulated in Eq. (37).

$$\Omega = \left\{ \left(\mathbf{P}_{\mathrm{PV}}^{\omega}, \pi^{\omega} \right) \middle| \sum_{\omega \in \Omega} \pi^{\omega} = 1, \pi^{\omega} \in (0,1] \right\} \tag{37}$$

where Ω denotes the typical PV scenario set, while $\mathbf{P}_{\mathrm{PV}}^{\omega}$, π^{ω} represent the PV profile and its probability under scenario ω. Accordingly, C_{BID}^{h} is a first-stage decision shared by all PV scenarios, whereas the scenario-dependent production schedule and $C_{\mathrm{ACT}}^{h,\omega}$ are second-stage recourse decisions.

For spot-price uncertainty, an IGDT uncertainty radius α is introduced around the baseline DA-RT price forecasts. The radius defines the admissible price-deviation set and provides the basis for evaluating the maximum price uncertainty that can be tolerated under a prescribed cost requirement. Thus, the DA-RT price uncertainty set is formulated in Eq. (38).

$$U(\alpha) = \left\{ \tilde{\lambda}^{h} \middle| \begin{array}{l} (1-\alpha)\lambda_{\mathrm{DA}}^{h} \le \tilde{\lambda}_{\mathrm{DA}}^{h} \le (1+\alpha)\lambda_{\mathrm{DA}}^{h} \\ (1-k_{\mathrm{RT}}\alpha)\lambda_{\mathrm{RT}}^{h} \le \tilde{\lambda}_{\mathrm{RT}}^{h} \le (1+k_{\mathrm{RT}}\alpha)\lambda_{\mathrm{RT}}^{h} \end{array}, \forall h \right\} \tag{38}$$

where $U(\alpha)$ denotes the DA-RT spot price sets, while $\tilde{\lambda}_{\mathrm{DA}}^{h}$, $\tilde{\lambda}_{\mathrm{RT}}^{h}$ represent the fluctuating prices under α. The coefficient k_{RT} indicates that RT volatility exceeds that of DA. Combining the settlement model and dual-uncertainty representations in Eqs. (29)-(38), the next subsection formulates the SC-IGDT co-optimization model for production scheduling and bidding.

3.2 SC-IGDT Co-Optimization Model for Production Scheduling and Bidding

Based on the above settlement mechanism and PV-price uncertainty modeling, an SC-IGDT co-optimization model for production scheduling and bidding is formulated. Its decision variables are collected as Eq. (39):

$$\boldsymbol{X} = \mathrm{col}\{\boldsymbol{X}_{\mathrm{phy}}, \mathbf{C}_{\mathrm{BID}}, \mathbf{C}_{\mathrm{ACT}}\} \tag{39}$$

where $\mathbf{C}_{\mathrm{BID}}$ and $\mathbf{C}_{\mathrm{ACT}}$ denote the DA bidding volumes and actual electricity consumption under different PV scenarios.

For scenario ω, the worst-case spot operating cost over the price uncertainty set $U(\alpha)$ is defined in Eq. (40):

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$$F_{\text{worst}}^{\omega}(\boldsymbol{X},\alpha)=\max_{\tilde{\boldsymbol{\lambda}}\in U(\alpha)}F_{\text{spot}}^{\omega}(\boldsymbol{X},\tilde{\boldsymbol{\lambda}})\quad\forall\omega \tag{40}$$

where F_{spot}^{ω} consists of the fixed MLT cost F_{MLT}^{h}, the spot exposure costs $F_{\text{DA,Dev}}^{h}$, $F_{\text{RT,Dev}}^{h,\omega}$, $F_{\text{PEN}}^{h,\omega}$, and the power-regulation costs $F_{\text{cpr}}^{h,\omega}$, $F_{\text{ud}}^{h,\omega}$. Accordingly, the SP-CVaR comprehensive risk cost J_{worst} is formulated as Eqs. (41)-(43):

$$J_{\text{worst}}=(1-\beta)\sum_{\omega\in\Omega}\pi^{\omega}F_{\text{worst}}^{\omega}+\beta\text{CVaR}_{\gamma}(F_{\text{worst}}) \tag{41}$$

$$\text{CVaR}_{\gamma}(F_{\text{worst}})=\zeta+\frac{1}{1-\gamma}\sum_{\omega\in\Omega}z^{\omega}\pi^{\omega}\quad\forall\omega \tag{42}$$

$$z^{\omega}\geq F_{\text{worst}}^{\omega}-\zeta,z^{\omega}\geq 0\quad\forall\omega \tag{43}$$

where β and γ denote the risk-aversion coefficient and CVaR confidence level. $\text{CVaR}_{\gamma}(F_{\text{worst}}),\zeta,z^{\omega}$ denote the CVaR, VaR, and excess tail-loss variable, respectively. The resulting cost serves as the IGDT performance measure, jointly accounting for PV scenario probabilities, tail risk, and price uncertainty.

Since the MLT contract cost remains fixed over the decision horizon and does not affect bidding or production scheduling, the optimal spot exposure cost at α=0, excluding the MLT cost, is adopted as the nominal benchmark in Eq. (44):

$$J_0=\min_{\boldsymbol{X},\zeta,z}[J_{\text{worst}}\big|_{\alpha=0}-\sum_{h\in H}F_{\text{MLT}}^{h}] \tag{44}$$

Given an acceptable cost increase δ, the SC-IGDT model maximizes the price uncertainty radius while restricting the spot exposure cost to $(1+\delta)J_0$, which is shown in Eq. (45):

$$\begin{aligned}&\max_{\boldsymbol{X},\alpha,\zeta,z}\quad\alpha\\&s.t.\quad J_{\text{worst}}-\sum_{h\in H}F_{\text{MLT}}^{h}\leq(1+\delta)J_0,\\&0\leq\alpha\leq\alpha_{\max},\text{Eqs.}(1)-(26),(28)-(44)\end{aligned} \tag{45}$$

where $\alpha_{\max}$ denotes the upper bound of the uncertainty-radius search interval. PV uncertainty is characterized by scenario probabilities and CVaR, whereas price uncertainty is represented by the IGDT radius. However, the continuous uncertainty set $U(\alpha)$ introduces infinitely many price-indexed constraints, resulting in a semi-infinite programming problem. Moreover, directly optimizing α generates bilinear terms involving C_{BID}^{h}, $C_{\text{ACT}}^{h,\omega}$. The next subsection therefore employs a joint-vertex affine-epigraph reformulation and an outer radius search to determine the optimal uncertainty radius α^{*}.

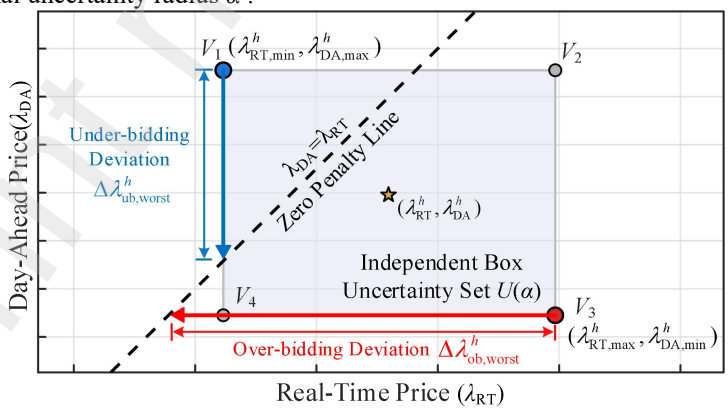


Fig. 3. Joint vertices and worst-case deviation spreads of the box-shaped DA-RT price uncertainty set.

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3.3 Joint-Vertex Affine-Epigraph Reformulation and Outer Bisection Algorithm

To address the SIP problem, a fixed price-uncertainty radius α is first considered. As indicated by Eqs. (31)-(32), the DA and RT deviation settlement costs are affine functions of prices, while the deviation-arbitrage penalty in Eq. (36) is the maximum of several affine functions. Therefore, for given bidding and scheduling decisions, the hourly spot settlement cost is convex and piecewise affine in the DA-RT prices.

As shown in Fig. 3, the DA-RT price intervals form a two-dimensional box uncertainty set, whose vertex set $V^h(\alpha) = \{V1, V2, V3, V4\}$ represents the combinations of the DA and RT price bounds. The dashed line denotes the zero-spread boundary. According to Eq. (36), the worst-case penalty search identifies the maximum positive deviation of a price point from the zero-spread line within this set, which is attained at a vertex. The red and blue segments represent the worst-case spreads under over-bidding and under-bidding.

The deviation settlement cost at vertex v is defined as Eq. (46). An epigraph variable $\theta^{h,\omega}$ is introduced to represent the maximum deviation settlement and arbitrage-recovery cost over the joint price vertices, shown in Eq. (47). Accordingly, the worst-case spot operating cost in Eq. (40) is reformulated as Eq. (48).

$$L^{h,\omega,v} = F^{h,v}_{\mathrm{DA,Dev}} + F^{h,\omega,v}_{\mathrm{RT,Dev}} \quad \forall v \in V^h(\alpha) \tag{46}$$

$$\begin{cases} \theta^{h,\omega} \geq L^{h,\omega,v} + \mu_{\mathrm{PEN}} \Delta\lambda^{h,v}_{\mathrm{ub}} [(1 - V_{\mathrm{PEN}}) C^{h,\omega}_{\mathrm{ACT}} - C^{h}_{\mathrm{BID}}] \\ \theta^{h,\omega} \geq L^{h,\omega,v} + \mu_{\mathrm{PEN}} \Delta\lambda^{h,v}_{\mathrm{ob}} [C^{h}_{\mathrm{BID}} - (1 + V_{\mathrm{PEN}}) C^{h,\omega}_{\mathrm{ACT}}] \quad \forall v \in V^h(\alpha) \\ \theta^{h,\omega} \geq L^{h,\omega,v} \end{cases} \tag{47}$$

$$F^{\omega}_{\mathrm{worst}} = \sum_{h \in H} (F^{h}_{\mathrm{MLT}} + F^{h,\omega}_{\mathrm{cpr}} + F^{h,\omega}_{\mathrm{ud}} + \theta^{h,\omega}) \quad \forall \omega \tag{48}$$

For fixed α, all vertex prices and corresponding price spreads are constants, and the above epigraph constraints are linear. Hence, the continuous worst-case price search is replaced by a finite set of joint-vertex constraints, converting the fixed-radius SIP into an equivalent MILP. The extended decision-variable vector, including the original scheduling and auxiliary variables, is defined as Eq. (49):

$$\tilde{\boldsymbol{X}} = \mathrm{col}\{\boldsymbol{X}, \boldsymbol{\theta}, \mathbf{F}_{\mathrm{worst}}, \mathbf{z}^{\mathrm{CVaR}}, \zeta\} \tag{49}$$

For a given radius α, the optimal value of the fixed-radius affine-epigraph MILP is defined as Eq. (50):

$$\Phi(\alpha) = \min_{\tilde{\boldsymbol{X}}} [J_{\mathrm{worst}}(\tilde{\boldsymbol{X}}; \alpha) - \sum_{h \in H} F^{h}_{\mathrm{MLT}}] \tag{50}$$

where $\Phi(\alpha)$ denotes the minimum worst-case spot-exposure risk cost attainable through optimal bidding and scheduling under radius α. If any decision satisfies the IGDT constraint, $\Phi(\alpha)$ cannot exceed its corresponding cost; conversely, the optimizer associated with $\Phi(\alpha)$ provides a feasible solution to the original problem. Therefore, the conventional SC-IGDT problem is equivalently expressed as Eq. (51):

$$\alpha^{*} = \max\{\alpha \in [0, \alpha_{\max}] : \Phi(\alpha) \leq (1 + \delta) J_0\} \tag{51}$$

Furthermore, increasing α expands the price uncertainty set; hence, the worst-case cost under any fixed decision cannot decrease. Algorithm 1 therefore applies outer bisection to determine the maximum feasible IGDT radius α^*.

Integrating the ThE-RTN with SC-IGDT, the framework jointly determines DA bids, production schedules, and the maximum feasible price-uncertainty radius, while joint-vertex affine-epigraph reformulation and outer bisection ensure tractability. The framework is illustrated in Fig. 4.

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Algorithm 1: Outer Bisection for the Maximum Feasible IGDT Radius

Input: Cost tolerance δ, search bound $\alpha_{\max}$, and radius tolerance ε_α
Output: Maximum feasible radius α^*
Set $J_0 \leftarrow \Phi(0)$, $\alpha_L \leftarrow 0$, $\alpha_U \leftarrow \alpha_{\max}$.
if $\Phi(\alpha_U) \le (1+\delta)J_0$ then
  return $\alpha^* \leftarrow \alpha_U$.
end if
while $\alpha_U - \alpha_L > \varepsilon_\alpha$ do
  Set $\bar{\alpha} \leftarrow (\alpha_L+\alpha_U)/2$.
  Solve the fixed-$\bar{\alpha}$ joint-vertex affine-epigraph MILP to obtain $\Phi(\bar{\alpha})$.
  if $\Phi(\bar{\alpha}) \le (1+\delta)J_0$ then
    Set $\alpha_L \leftarrow \bar{\alpha}$.
  else
    Set $\alpha_U \leftarrow \bar{\alpha}$.
  end if
end while
return $\alpha^* \leftarrow \alpha_L$.

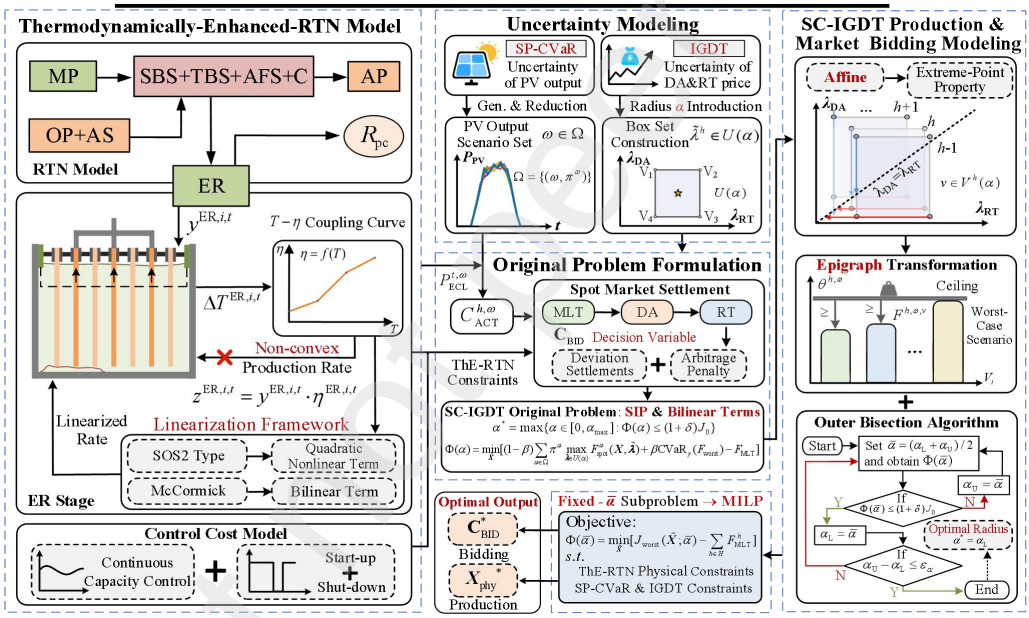


Fig. 4. Schematic of the coordinated production-scheduling and bidding framework.

4. Case Study

To validate the proposed coordinated production-scheduling and bidding framework, a case study based on an actual ECL in Yunnan, China, is conducted. The model is formulated in MATLAB R2023a via YALMIP and solved by Gurobi 13.0 (MIP gap: 0.01%) on an AMD Ryzen 9 9955HX CPU with 32 GB RAM. The 24-hour production cycle is discretized into 96 periods with a 15-minute time step.

4.1 Parameter Settings

The ECL features a 22 MW rated capacity and a 3 MW PV installation. Except for the ER and AS processes (comprising three series each), all others operate singularly; detailed parameters are listed in Table

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1. Initially, inventories for materials ia, ie, and io are 1000, 10, and 2400 (t or m^3), with others initialized to zero. Material boundary constraints refer to [13], with a 24-h production target of 7690 t. The electrolyte temperature is constrained within 65-70°C. The three ER series share identical derived thermodynamic coefficients, with A^{ER}=0.02711125, E^{ER}=0.01, F^{ER}=0.26288875, calculated from the thermodynamic parameters in Eq. (18). The SOS2 approximation uses 3 uniformly spaced breakpoints over [0.75, 1.05], with h=0.15.

Figs. 5 and 6 depict the ECL load baseline, the PV scenario set for the trading day, and the MLT and forecasted DA/RT baseline prices. Market and risk parameters are configured as follows: the MLT volume ratio α_{MLT} is 0.85; the arbitrage recovery μ_{PEN} and penalty-free ratios V_{PEN} are 0.8 and 0.1, respectively; the RT volatility coefficient k_{RT} is 1.2; and the CVaR weight β and confidence level γ are 0.2 and 0.9.

Table 1. Operational parameters of ECL processes.

Stage	Power (kW)	Boundary	Control Cost (CNY)			
			$\eta_{up}^{k,i}$	$\eta_{down}^{k,i}$	$\eta_{su}^{k,i}$	$\eta_{sd}^{k,i}$
MP	1579.3	[0.85,1.05]	0.021	0.01	10	20
SBS	1643.7	/	/	/	/	/
TBS	1643.7	/	/	/	/	/
AFS	1029.6	/	/	/	/	/
C	1110	/	/	/	/	/
ER_1	2168.9	[0.75,1.05]	0.03	0.014	20	40
ER_2	2168.9	[0.75,1.05]	0.03	0.014	20	40
ER_3	2168.9	[0.75,1.05]	0.03	0.014	20	40
OP	4556	[0.8,1.05]	0.021	0.011	/	/
AS_1	535.6	[0.95,1]	0.003	0.003	/	/
AS_2	465.6	[0.9,1]	0.003	0.003	/	/
AS_3	345.6	[0.8,1]	0.003	0.003	/	/
AP	2590.2	[0.9,1.05]	0.036	0.018	/	/

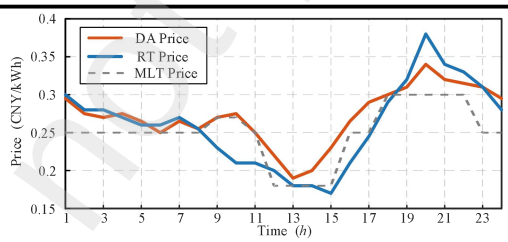


Fig. 5. MLT, DA, and RT price curves.

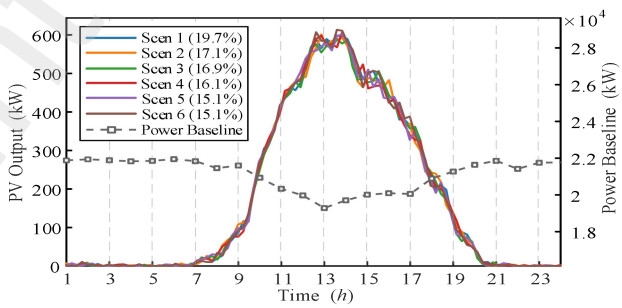


Fig. 6. ECL load baseline and typical PV output scenarios.

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*4.2 Determination and Interpretation of the Optimal IGDT Radius α**

With the cost tolerance set to δ=0.1, the nominal spot-exposure cost is J_0=16,475.20 CNY, yielding a cost limit of 18,122.72 CNY. As reported in Table 2, the final bisection stage converges after eight iterations and obtains α^*=3.781% within 148.9 s. At this radius, $\Phi(\alpha^*)$=18,122.15 CNY satisfies the prescribed cost limit, whereas the adjacent upper bound α_U=3.789% gives $\Phi(\alpha_U)$=18,124.68 CNY and becomes infeasible. Therefore, α^* is the maximum feasible price-uncertainty radius within the specified bisection tolerance.

Table 2. Determination of the maximum feasible IGDT radius.

Metric	Value
α_{max} (%)	50
ε_α	1×10^{-4}
J_0 (CNY)	16475.20
$(1+\delta)J_0$ (CNY)	18122.72
α^* (%)	3.781
$\Phi(\alpha^*)$ (CNY)	18122.15
α_U (%)	3.789
$\Phi(\alpha_U)$ (CNY)	18124.68

Fig. 7 further illustrates the price uncertainty bands and corresponding bidding decision. The bid generally remains near the expected net consumption when deviation exposure dominates. In contrast, substantial downward bid adjustments, including zero bids, arise when favorable DA-RT price spreads persist throughout the uncertainty range. This contrast reflects the trade-off between deviation exposure and persistent settlement incentives under the prescribed cost budget. The zero-bidding decisions are therefore examined through their marginal settlement costs. For the zero-bidding periods, the lower DA-price bound remains above the upper RT-price bound under α^*. Within the penalty-free region, the marginal settlement cost equals $\lambda_{DA}^{h,v}$-$\lambda_{RT}^{h,v}$. Once the bid falls below the penalty-free lower bound, it becomes Eq. (52).

$$\nabla F = \partial F^{h,\omega,v} / \partial C_{BID}^{h} = (1-\mu_{PEN})(\lambda_{DA}^{h,v} - \lambda_{RT}^{h,v}) \tag{52}$$

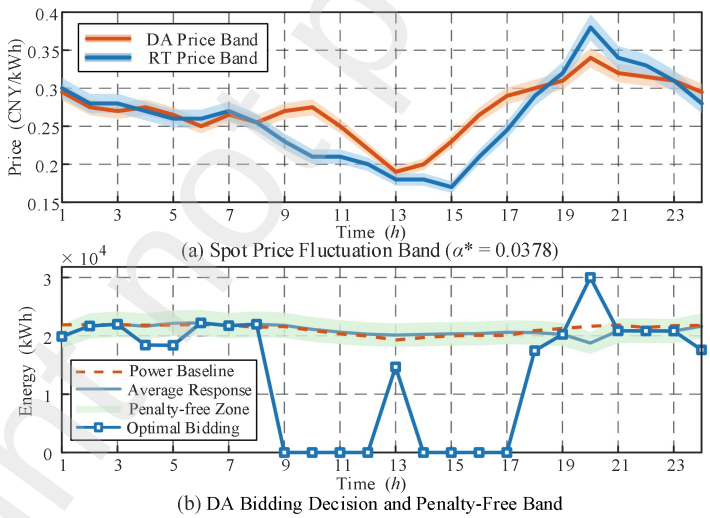


Fig. 7. Price uncertainty bands and the corresponding bidding decision.

This expression applies when C_{BID}^{h}<(1-V_{PEN})$C_{ACT}^{h,\omega}$; crossing C_{MLT}^{h} only changes the ECL from a DA buyer to a seller. Since gradients remain positive at all joint price vertices, reducing the DA bid decreases the worst-case settlement cost despite the arbitrage-recovery charge. For example, at h = 10, the DA-price lower bound still exceeds the RT-price upper bound by 0.0451 CNY/kWh under α^*. Hence, the optimal bid reaches its lower bound of zero. Conversely, at h=20, the RT-price lower bound exceeds the DA-price upper bound

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under α^*. The marginal cost remains negative at all joint vertices, so increasing the DA bid reduces the worst-case settlement cost despite arbitrage recovery, driving the bid beyond penalty-free band. Together, these decisions show that SC-IGDT does not impose uniformly conservative bidding; instead, it adjusts the bidding direction according to persistent DA-RT price spreads, thereby preserving favorable settlement opportunities while satisfying the prescribed cost budget.

To interpret the cost-tolerance parameter, four values of δ are considered, with the resulting maximum tolerable price deviations reported in Table 3. Increasing δ from 0.04 to 0.15 raises α^* from 1.188% to 6.516%. Their nonlinear relationship reflects the combined effects of production constraints and piecewise settlement rules, providing ECL operators with a direct reference for selecting the price-risk protection level according to their acceptable cost budget.

Table 3. Maximum tolerable price deviations.

δ	$(1+\delta)\,J_0$ (CNY)	α^* (%)
0.04	17134.21	1.188
0.08	17793.22	2.781
0.1	18122.72	3.781
0.15	18946.48	6.516

4.3 Effectiveness of Uncertainty Management

To evaluate price-uncertainty treatments, Table 4 compares four methods under identical ThE-RTN constraints, PV scenarios, CVaR settings, and settlement rules, with a budget exceedance uniformly defined as an OOS spot-exposure cost above $(1+\delta)J_0$. SC-RO uses α_{RO}=12.263% from the 95th percentile of daily maximum normalized training errors; SC-DRO uses eight representative error trajectories and a 95%-confidence probability box with ε_q=0.13426; and SC-IGDT uses δ=0.1, yielding α^*=3.781%. SC-RO and SC-DRO are calibrated on 160 synthetic trajectories, and all methods are evaluated on 5000 independent OOS trajectories.

Table 4. Out-of-sample performance of different price-uncertainty treatments.

Method	Nominal SC-CVaR	SC-RO	SC-DRO	SC-IGDT
Nominal cost (CNY)	16475	17789	16499	16911
Nominal cost premium (%)	0	7.98	0.15	2.65
OOS mean (CNY)	16596	17789	16534	16914
OOS $CVaR_{0.95}$ (CNY)	17263	17988	17017	17189
OOS P99 (CNY)	17407	18008	17140	17224
OOS maximum (CNY)	19114	18164	17618	17593
Budget exceedances (/5000)	2	3	0	0

At the original price-error scale, Table 4 shows that SC-DRO achieves the lowest OOS mean and $CVaR_{0.95}$. SC-RO incurs the largest nominal cost premium of 7.98%, reflecting the conservatism of its calibrated uncertainty set, whereas SC-IGDT requires a moderate premium of 2.65%. Compared with Nominal SC-CVaR, SC-IGDT reduces the observed maximum cost by 7.96% and, like SC-DRO, records no budget exceedances among 5000 OOS trajectories. These results indicate that SC-IGDT improves protection against sampled extreme costs at a moderate nominal cost premium.

To assess sensitivity to larger price deviations, the OOS price-error scale factor is increased from 1.0 to 2.0. Fig. 8 reports the increases in $CVaR_{0.95}$ and P99 relative to each method's results at scale 1.0, while Table 5 summarizes performance under doubled errors. The optimized decisions and calibration parameters remain fixed, and all methods are evaluated using the same 5000 OOS trajectories and budget threshold.

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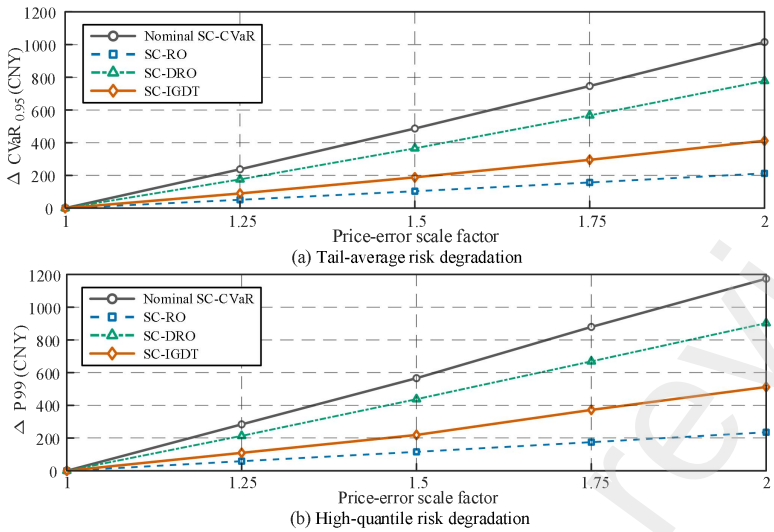


Fig. 8. Increases in out-of-sample tail-risk measures with increasing price-error magnitude: (a)$CVaR_{0.95}$; (b)P99.

Table 5. Out-of-sample performance under doubled price-error magnitude.

Method	Nominal SC-CVaR	SC-RO	SC-DRO	SC-IGDT
OOS mean (CNY)	16837	17792	16667	16931
Mean increase (CNY)	240.4	3.1	133.5	17.8
OOS $CVaR_{0.95}$ (CNY)	18278	18201	17794	17601
OOS P99 (CNY)	18581	18244	18043	17737
OOS maximum (CNY)	21922	18962	19031	18722
Budget exceedances (/5000)	132	229	41	16

As the error scale increases, SC-RO exhibits the smallest increases in $CVaR_{0.95}$ and P99, while SC-IGDT shows smaller increases than Nominal SC-CVaR and SC-DRO. Under doubled errors, SC-IGDT achieves the lowest absolute $CVaR_{0.95}$, P99, and observed maximum cost, with 16 budget exceedances among 5000 trajectories, compared with 41 for SC-DRO. Although SC-DRO retains a lower mean cost, SC-IGDT provides stronger tail-risk protection in this stress test at a lower nominal cost premium than SC-RO. Moreover, IGDT determines the maximum admissible price-deviation radius under a prescribed cost tolerance and uncertainty-set structure without specifying a price-error probability distribution. These results support its suitability for balancing operating economy and adverse-price protection in the present scheduling and bidding problem.

For PV uncertainty management, owing to limited PV penetration and scenario dispersion, varying β from 0 to 0.8 changes α^* by less than 10^{-4}, indicating that price uncertainty dominates the present risk profile. Hence, CVaR mainly serves as an auxiliary safeguard against low-probability PV deviations rather than a primary driver of the current bidding decision.

4.4 Operational Strategy Comparison and Economic Performance

To quantify the incremental economic contributions of the ThE-RTN and SC-IGDT, Table 6 compares three operational strategies under the same case settings and uncertainty radius α^*. Strategy 1 employs self-scheduled conventional RTN with conventional bidding; Strategy 2 employs price-responsive ThE-RTN with conventional bidding; and Strategy 3 adopts the coordinated production-scheduling and bidding framework. Since the MLT cost is fixed across strategies, only the spot-exposure cost components and risk cost are reported. A negative DA deviation cost denotes a settlement credit relative to the MLT position.

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Table 6. Cost decomposition and spot-exposure risk cost of three operating strategies.

Metric	Strategy 1	Strategy 2	Strategy 3
Expected $F_{\mathrm{DA,Dev}}$ (CNY)	22011	19704	-21925
Expected $F_{\mathrm{RT,Dev}}$ (CNY)	2.86	3.12	36843
Expected F_{PEN} (CNY)	0	0	3000.2
Expected $F_{\mathrm{cpr}}+F_{\mathrm{ud}}$ (CNY)	19.33	168.19	197.56
$J_{\mathrm{worst}}-\sum F_{\mathrm{MLT}}$ (CNY)	22039	19877	18122
Cost Reduction Rate (%)	/	9.81	17.77

As shown in Table 6, Strategy 2 reduces risk cost from 22,039 to 19,877 CNY, corresponding to a 9.81% decrease relative to Strategy 1. Although the internal-control cost increases from 19.33 to 168.19 CNY, the expected DA deviation cost decreases by 2307 CNY. This indicates that ThE-RTN-based price-responsive scheduling exploits thermodynamically feasible flexibility to reduce market exposure at a limited control cost. Strategy 3 further reduces risk cost to 18,122 CNY, providing an additional saving of 8.83%. Although RT deviation and arbitrage-recovery costs increase, they are more than offset by the DA settlement credit. Hence, ThE-RTN-based scheduling already realizes part of the economic value of physical flexibility, while coordinating production with DA bidding and DA-RT settlement decisions further improves this value. Overall, the proposed strategy reduces risk cost by 17.77% relative to Strategy 1, demonstrating that the ThE-RTN enables executable price-responsive flexibility, while SC-IGDT further realizes its economic value through coordinated production scheduling and bidding.

4.5 Physical Feasibility and Operational Safety of the ThE-RTN

To verify the physical feasibility and operational safety of the ThE-RTN, Figs. 9-11 demonstrate its coordinated power regulation, material and production balance, and temperature-limit compliance under the most probable PV scenario (ω=1), with the conventional RTN schedule thermodynamically replayed for comparison.

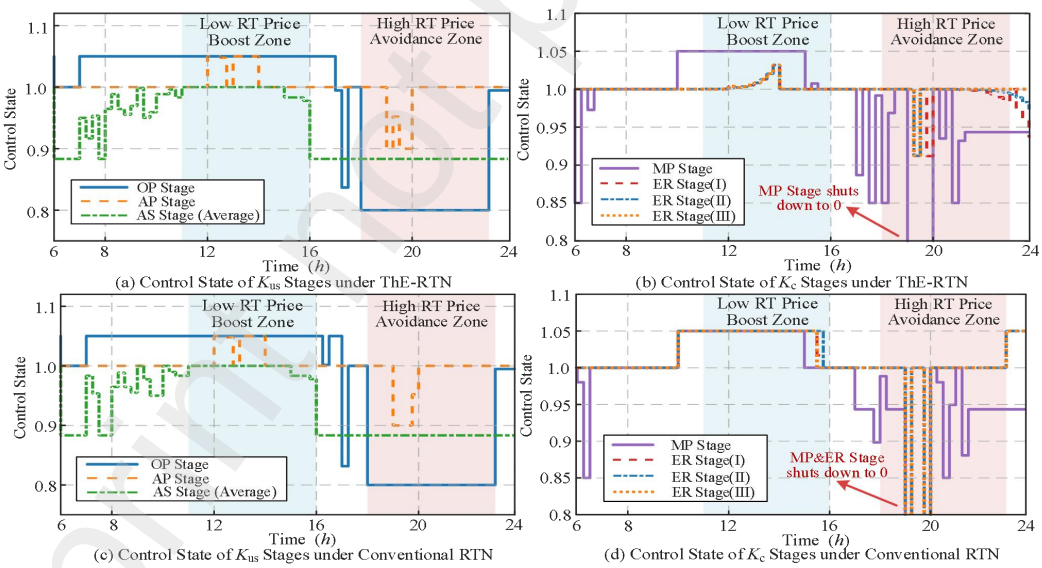


Fig. 9. Power-control states of the main ECL stages under (a)-(b) the ThE-RTN and (c)-(d) the conventional RTN.

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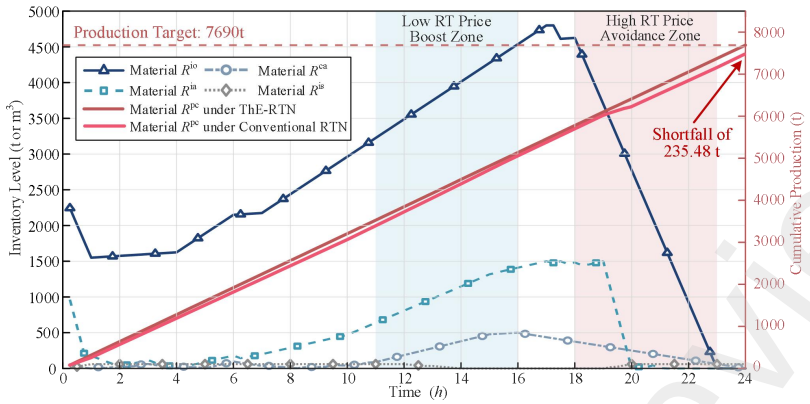


Fig. 10. Material-inventory trajectories under ThE-RTN and cumulative production obtained by thermodynamic replay of the two RTN schedules.

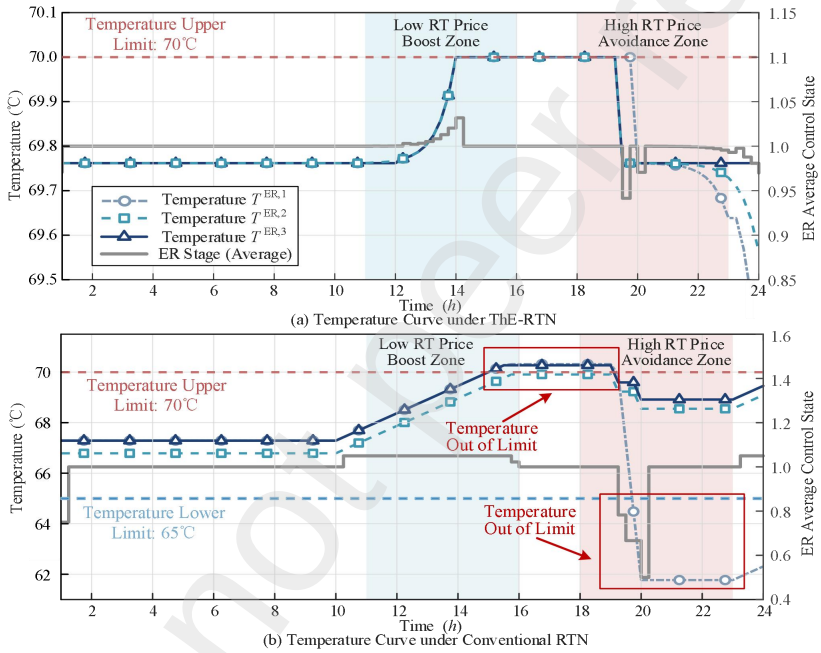


Fig. 11. Electrolyte-temperature trajectories and average ER control states under (a) the ThE-RTN and (b) the conventional RTN.

As shown in Figs. 9 and 10, both models shift load toward low-price periods, but the conventional RTN concentrates deeper curtailments and equipment shutdowns during high-price hours. The ThE-RTN instead coordinates power regulation with material buffers, accumulating inventory before the high-price interval and sustaining downstream production during curtailment. Under thermodynamic replay, the prolonged low-temperature operation induced by the conventional schedule causes cumulative efficiency losses, yielding only 7454.52 t and a production shortfall of 235.48 t.

Fig. 11 explains this production gap. The ThE-RTN maintains the ER temperatures within 65-70 °C, whereas the conventional RTN drives them above the upper limit during low-price boosting and below the lower limit following concentrated curtailment. These violations demonstrate that the conventional RTN overestimates executable flexibility, while the ThE-RTN preserves thermodynamic feasibility and production reliability. Overall, the ThE-RTN converts nominal scheduling flexibility into thermodynamically feasible and production-reliable flexibility.

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Finally, the accuracy of the SOS2-McCormick reformulation is evaluated by comparing the approximation error revealed by nonlinear replay with the theoretical upper bound in Eq. (27). The original SC-IGDT plan yields a terminal production of 7690 t in the MILP model, whereas replaying the same plan using the original nonlinear relationships gives a minimum terminal production of 7675.19 t across all representative PV scenarios. The resulting maximum shortfall is 14.81 t, corresponding to a relative discrepancy of 0.1926%, which is below the theoretical error bound of 0.3954%. Although the replay error remains within the theoretical bound, the residual production shortfall necessitates further compensation to ensure physical executability at the specified production precision.

For nonlinear-replay validation, a terminal-production tolerance of 1 t is predefined, accounting for approximately 0.013% of the 7690 t production target. Accordingly, a nonlinear-replay terminal production of at least 7689 t is regarded as satisfying the production requirement at this precision. For each PV scenario, the difference between the terminal productions obtained from the linear model and nonlinear replay is calculated, and the portion exceeding the prescribed tolerance is added to the corresponding linear terminal-production requirement. The DA bid and production schedule are subsequently reoptimized, with all other model parameters and uncertainty settings unchanged. This replay-and-compensation experiment is conducted independently of the primary economic comparison. Table 7 summarizes the results.

Table 7. Nonlinear replay and terminal-production compensation results.

Metric	SC-IGDT Plan	Compensated Plan
Minimum nonlinear terminal production (t)	7675.19	7689.68
Maximum terminal shortfall (t)	14.810	0.321
Maximum relative shortfall (%)	0.1926	0.0042
Scenarios satisfying the production tolerance	0/6	6/6
Temperature/material/control feasibility	Feasible	Feasible

After re-optimization, the minimum terminal production obtained from nonlinear replay across all scenarios increases to 7689.68 t, while the maximum residual shortfall decreases from 14.81 to 0.321 t, below the prescribed 1-t tolerance. Meanwhile, all temperature, material, and control constraints remain satisfied. The corresponding risk cost increases from 18122.15 CNY to 18206.18 CNY, an increase of only 84.02 CNY, or 0.464%. These results demonstrate that accounting for the linearization-induced production deviation restores production feasibility within the specified tolerance and preserves physical feasibility at a limited additional cost, further confirming the effectiveness of the proposed framework in coordinating economic decisions with physical execution.

5. Conclusion

Under high renewable penetration, grid-interactive PILs must balance economic performance, uncertainty protection, and physical feasibility under PV and price uncertainties. Addressing thermodynamic neglect and decoupled scheduling-risk decisions, we develop a coordinated production-scheduling and bidding framework. ThE-RTN embeds power-temperature-efficiency coupling into plant-wide scheduling, while SC-IGDT manages scenario-based PV and DA-RT price risks under settlement rules. Case studies show that a 10% cost tolerance yields a maximum feasible price-uncertainty radius of 3.781%. SC-IGDT further mitigates extreme costs and reduces budget exceedances under amplified price errors. The proposed framework reduces spot-exposure risk cost by 17.77% relative to conventional RTN scheduling and bidding. Thermodynamic replay verifies production and thermal feasibility, whereas conventional RTN causes thermal violations and a production shortfall. These results demonstrate coordinated improvements in uncertainty protection, economic performance, and operational safety. Future work will investigate intraday rolling optimization and control.

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Coordinated Thermodynamically Feasible Production Scheduling and Bidding for Process Industrial Loads Under PV and Price Uncertainties

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HIGHLIGHTS

- Thermodynamic constraints preserve production feasibility in industrial scheduling.
- SOS2-McCormick approximation enables tractable thermodynamic scheduling.
- SC-IGDT coordinates production and bidding under PV and price uncertainties.
- The framework improves economic performance while maintaining production safety.

ABSTRACT

High renewable-energy penetration increases net-load variability and spot-price uncertainty, strengthening the need for flexible industrial demand in electricity markets. However, for process industrial loads (PILs), uncertain photovoltaic (PV) output and electricity prices interact with tightly coupled material and thermodynamic dynamics, linking economic bidding to production feasibility and operational safety. This paper proposes a coordinated production-scheduling and bidding framework for an electrolytic copper load by integrating a thermodynamically enhanced resource-task network (ThE-RTN) with stochastic programming-CVaR-information gap decision theory (SC-IGDT). The ThE-RTN embeds power-temperature-efficiency coupling into plant-wide material-flow scheduling and is reformulated via SOS2-McCormick linearization. SC-IGDT combines scenario-based CVaR for PV uncertainty with IGDT for bounded day-ahead and real-time price uncertainty under settlement and arbitrage-recovery mechanisms. A joint-vertex affine-epigraph reformulation with outer bisection determines the maximum feasible uncertainty radius. Case studies show that a 10% cost tolerance yields a maximum feasible price-uncertainty radius of 3.781%, and the proposed framework reduces spot-exposure risk cost by 17.77% relative to conventional RTN scheduling and bidding. SC-IGDT further provides effective protection against extreme costs and budget violations. Thermodynamic replay confirms production and temperature feasibility, whereas the conventional RTN causes thermal violations and a production shortfall. These results demonstrate effective coordination among economic efficiency, uncertainty protection, and operational safety.

Keywords: Process industrial load (PIL); photovoltaic and price uncertainties; resource-task network (RTN); information gap decision theory (IGDT); production scheduling and bidding.

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1. Introduction

1.1 Background and Motivation

Under the dual-carbon goals and high renewable penetration, renewable variability has increased the need for demand-side flexibility in power grids [1]-[2]. Process industrial loads (PILs) are promising large-scale flexibility resources because of their substantial electricity demand, strong power-regulation capability, material buffers, and thermal inertia [3]-[5]. Market-based demand-response scheduling can mobilize this flexibility by coordinating production schedules with electricity consumption, thereby compensating for renewable variability while supporting secure grid operation [6]. Yet feasible PIL responses cannot be determined from electrical limits alone; they also depend on production continuity, intermediate-material inventories, equipment operating conditions, and the carryover of internal process states across periods [7]-[8]. Economically attractive power adjustments must also respect process dynamics and safe operating limits to be implementable in practice [9]. Accordingly, plant-wide interstage material flows and power-temperature-efficiency dynamics must be represented to translate potential flexibility into operationally feasible and reliable grid support.

1.2 Literature Review and Research Gap

State-Task Network (STN) and Resource-Task Network (RTN) models are widely used in PIL scheduling to represent production tasks, equipment states, material inventories, and inter-stage couplings. Representative applications of process-level scheduling include coordinating steelmaking production with energy and reserve-market bidding [10], and integrating cement production with on-site renewables, storage, and electricity-market participation [11]. Recent work has improved the scalability of RTN-based industrial scheduling through continuous task representations and unified constraint formulations [12]. However, when process behavior is represented mainly by fixed conversion relationships or predefined operating modes, internal-state evolution and its intertemporal effects under power regulation may be insufficiently captured [13]-[14]. Recent studies have incorporated process-state dynamics into STN models, but the resulting nonconvex mixed-integer formulations still require tailored algorithms [15]. Meanwhile, thermodynamics-aware scheduling models have begun to consider temperature-dependent efficiency and safety limits [16]-[18], yet generally focus on specific devices or local processes. Integrating power-temperature-efficiency coupling into a plant-wide scheduling framework with multi-stage material networks while retaining industrial-scale tractability remains an open challenge.

Beyond physical modeling, grid-interactive PILs with distributed photovoltaic (PV) generation also face uncertainties in PV output and spot prices. PV forecast errors can generally be represented by probabilistic scenarios derived from historical samples, whereas spot prices exhibit spikes and nonstationarity, making stable distributions difficult to identify [19]-[20]. In day-ahead-real-time (DA-RT) markets with deviation settlement and arbitrage-recovery rules, these uncertainties may further translate into settlement risk. Existing studies use scenario-based stochastic programming (SP), conditional value-at-risk (CVaR), robust optimization (RO), distributionally robust optimization (DRO), and information-gap decision theory (IGDT) under different information conditions [19],[21]. Applying RO to both uncertainties cannot fully exploit probabilistic PV information, while DRO depends on samples and ambiguity-set construction. By contrast, IGDT evaluates tolerable price deviations under a prescribed cost requirement. Thus, combining SP/CVaR for PV uncertainty with IGDT for price uncertainty better matches their distinct characteristics. A recent study applies CVaR and entropy-weighted IGDT to multi-stage VPP trading in coupled electricity-green markets [22]. However, risk-based market decisions remain insufficiently coordinated with the physical production feasibility of PILs. How to jointly address heterogeneous PV and price uncertainties and complex settlement rules while coordinating production scheduling, market bidding, and settlement risk for the safe, reliable, and economic utilization of PIL flexibility remains an open issue.

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1.3 Contribution and Summary

In summary, the participation of PILs in renewable-integrated grid interactions requires both an accurate characterization of production-constrained executable flexibility and its reliable and economic market utilization under PV and spot-price uncertainties. Using an actual electrolytic copper load (ECL) in Yunnan as the study system, this paper develops a coordinated production-scheduling and bidding optimization method combining a Thermodynamically Enhanced Resource-Task Network (ThE-RTN) with Stochastic Programming-CVaR-Information-Gap Decision Theory (SC-IGDT). The main contributions are as follows:

• A ThE-RTN model is developed by incorporating the power-temperature-efficiency relationship of electrolysis into the RTN, thereby capturing the cross-period effects of power regulation on thermal states, production efficiency, and yield. The resulting nonconvex couplings are handled through a Type 2 (SOS2)-McCormick formulation, yielding an MILP-solvable production scheduling model.

• An SC-IGDT production-scheduling and bidding co-optimization model is formulated for heterogeneous uncertainties of PV and DA-RT prices, while accounting for deviation settlement and arbitrage-recovery mechanisms. Based on the ThE-RTN, the model jointly optimizes DA bids and production schedules and determines the maximum tolerable price deviation under a prescribed spot-exposure cost tolerance.

• To address the SIP induced by the continuous price set and the bilinear coupling between the uncertainty radius and the bidding and actual net-purchase variables, a joint-vertex affine-epigraph reformulation and an outer bisection algorithm are developed. The fixed-radius subproblem is converted into an MILP, yielding the maximum feasible price-uncertainty radius and its associated production schedule and DA bids.

The remainder of this paper is organized as follows. Section 2 establishes the ThE-RTN model and its SOS2-McCormick approximation for the ECL. Section 3 formulates the SC-IGDT co-optimization model for production scheduling and bidding and presents the joint-vertex affine-epigraph reformulation and outer bisection algorithm. Section 4 presents the case studies, and Section 5 concludes the paper.

2. Thermodynamically Enhanced RTN Scheduling Model for PILs

Conventional RTN models cannot fully capture the cross-period effects of power regulation on thermal states, efficiency, and production. Therefore, a ThE-RTN is developed for the ECL by embedding power-temperature-efficiency dynamics into its multi-stage material network, while a hybrid SOS2-McCormick formulation is used to handle the resulting nonconvexities.

The model assumes that: 1) power regulation states change only at interval boundaries, with inter-stage material transfer delays negligible relative to the scheduling interval; and 2) ambient temperature remains constant over the scheduling horizon, while the narrow electrolyte temperature range allows heat-dissipation power to be approximated as constant.

2.1 Process Characteristic Analysis and Power Regulation Cost Modeling of ECL

As a typical PIL, the ECL integrates sequential pyrometallurgical processes and auxiliary units. Raw materials from Material Preparation (MP) undergo progressive smelting-Side-Blowing (SBS), Top-Blowing (TBS), and Anode Furnace Smelting (AFS)-followed by Casting (C) to yield anode copper, which is finally electrolyzed via Electrolytic Refining (ER) into cathode copper. Concurrently, Auxiliary Smelting (AS) and Oxygen Production (OP) supply pyrometallurgical oxygen, while Acid Production (AP) recovers smelting gases.

Power regulation methods for these stages are typically categorized into continuous power regulation and discrete load shedding. Since ECL production scheduling decisions must account for PV output characteristics, the power and regulation characteristics of the equipment under PV scenarios can be formulated as Eqs. (1)-(2) [13]:

$$P^{k,i,t,\omega} = P^{k,i}_{\text{rate}} x^{k,i,t,\omega} s^{k,i,t,\omega} = P^{k,i}_{\text{rate}} y^{k,i,t,\omega} \quad \forall k,i,t,\omega \tag{1}$$

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$$\begin{cases} \underline{x}^{k,i} \le x^{k,i,t,\omega} \le \overline{x}^{k,i}, s^{k,i,t,\omega} \in \{0,1\} & k \in K_{\mathrm{c}} \\ \underline{x}^{k,i} \le x^{k,i,t,\omega} \le \overline{x}^{k,i}, s^{k,i,t,\omega} = 1 & k \in K_{\mathrm{us}} \\ x^{k,i,t,\omega} = s^{k,i,t,\omega} = 1 & k \in K_{\mathrm{uc}} \end{cases} \tag{2}$$

where $P^{k,i,t,\omega}$, $P^{k,i}_{\mathrm{rate}}$, $x^{k,i,t,\omega}$, $s^{k,i,t,\omega}$, $y^{k,i,t,\omega}$denote the equipment power, rated power, continuous power regulation state, commitment status, and overall power regulation state, respectively. k, i, t, ω represent the indices for the process stage, equipment, 15-minute time interval, and PV output scenario, respectively. $\underline{x}^{k,i}, \overline{x}^{k,i}$ denotes the power regulation boundaries. Additionally, K_{c}={MP,ER} and K_{us}={OP,AS,AP} denote the sets of controllable and non-sheddable stages, respectively, while the remaining uncontrollable stages belong to the set K_{uc}. Eq. (1) indicates that power consumption depends on distinct power regulation methods, while Eq. (2) captures the power regulation characteristics of different process stages. Since the discrete variables in Eq. (1) render the model non-convex, a new variable $y^{k,i,t,\omega}$ is introduced to represent the overall power regulation state of the equipment, which is then linearized via the tight Big-M method, as formulated in Eq. (3):

$$\begin{cases} y^{k,i,t,\omega} \le \overline{x}^{k,i} s^{k,i,t,\omega} \\ y^{k,i,t,\omega} \le x^{k,i,t,\omega} - \underline{x}^{k,i}(1 - s^{k,i,t,\omega}) \\ y^{k,i,t,\omega} \ge x^{k,i,t,\omega} - \overline{x}^{k,i}(1 - s^{k,i,t,\omega}) \\ y^{k,i,t,\omega} \ge \underline{x}^{k,i} s^{k,i,t,\omega} \end{cases} \quad \forall k,i,t,\omega \tag{3}$$

The scheduling cost induced by power regulation is the core factor constraining the optimal scheduling strategy of the ECL. It comprises continuous regulation capacity costs and discrete start-up/shut-down costs, formulated as Eqs. (4)-(9):

$$F^{h,\omega}_{\mathrm{cpr}} = \sum_{t \in T_h} \sum_{k \in K} \sum_{i \in I_k} P^{k,i}_{\mathrm{rate}} (\Delta y^{k,i,t,\omega}_{\mathrm{up}} \eta^{k,i}_{\mathrm{up}} + \Delta y^{k,i,t,\omega}_{\mathrm{down}} \eta^{k,i}_{\mathrm{down}}) \quad \forall h,\omega \tag{4}$$

$$F^{h,\omega}_{\mathrm{ud}} = \sum_{t \in T_h} \sum_{k \in K} \sum_{i \in I_k} (u^{k,i,t,\omega} \eta^{k,i}_{\mathrm{su}} + d^{k,i,t,\omega} \eta^{k,i}_{\mathrm{sd}}) \quad \forall h,\omega \tag{5}$$

$$y^{k,i,t,\omega} - 1 = \Delta y^{k,i,t,\omega}_{\mathrm{up}} - \Delta y^{k,i,t,\omega}_{\mathrm{down}} \quad \forall k,i,t,\omega \tag{6}$$

$$\Delta y^{k,i,t,\omega}_{\mathrm{up}}, \Delta y^{k,i,t,\omega}_{\mathrm{down}} \ge 0 \quad \forall k,i,t,\omega \tag{7}$$

$$u^{k,i,t,\omega} - d^{k,i,t,\omega} = s^{k,i,t,\omega} - s^{k,i,t-1,\omega} \quad \forall k,i,t,\omega \tag{8}$$

$$u^{k,i,t,\omega} + d^{k,i,t,\omega} \le 1 \quad \forall k,i,t,\omega \tag{9}$$

where $F^{h,\omega}_{\mathrm{cpr}}$, $F^{h,\omega}_{\mathrm{ud}}$ denote the continuous power regulation cost and discrete start-up/shut-down cost, respectively. h, T_h, K, I_k represent the 1-hour time interval index, the set of time periods t within hour h, the set of process stages, and the set of units i within stage k. $\Delta y^{k,i,t,\omega}_{\mathrm{up}}$, $\Delta y^{k,i,t,\omega}_{\mathrm{down}}$ denote the continuous upward and downward regulation amounts, with $\eta^{k,i}_{\mathrm{up}}$, $\eta^{k,i}_{\mathrm{down}}$ being their unit costs. $\eta^{k,i}_{\mathrm{su}}$, $\eta^{k,i}_{\mathrm{sd}}$ represent the equipment start-up and shut-down costs, while $u^{k,i,t,\omega}$, $d^{k,i,t,\omega}$ indicate the corresponding start-up and shut-down states. Eqs. (6)-(7) calculate the specific upward and downward regulation capacities within time period t. Eq. (8) determines the start-up and shut-down states, while Eq. (9) prevents simultaneous start-up and shut-down actions.

While Eqs. (1)-(9) quantify ECL regulation and costs, defining actual operational bounds requires inter-stage material and intra-stage thermodynamic couplings. Accordingly, the next section embeds these into the conventional RTN to formulate the ThE-RTN model.

2.2 Formulation of the Enhanced RTN Scheduling Model Incorporating Thermodynamic Coupling Mechanisms

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Material interactions in the ECL are constrained by their physical states. For instance, molten materials in pyrometallurgical stages cannot be exchanged across parallel production lines. Accordingly, the interaction dynamics for different materials are formulated in Eqs. (10)-(12), and the conventional RTN model for the ECL is illustrated in Fig. 1.

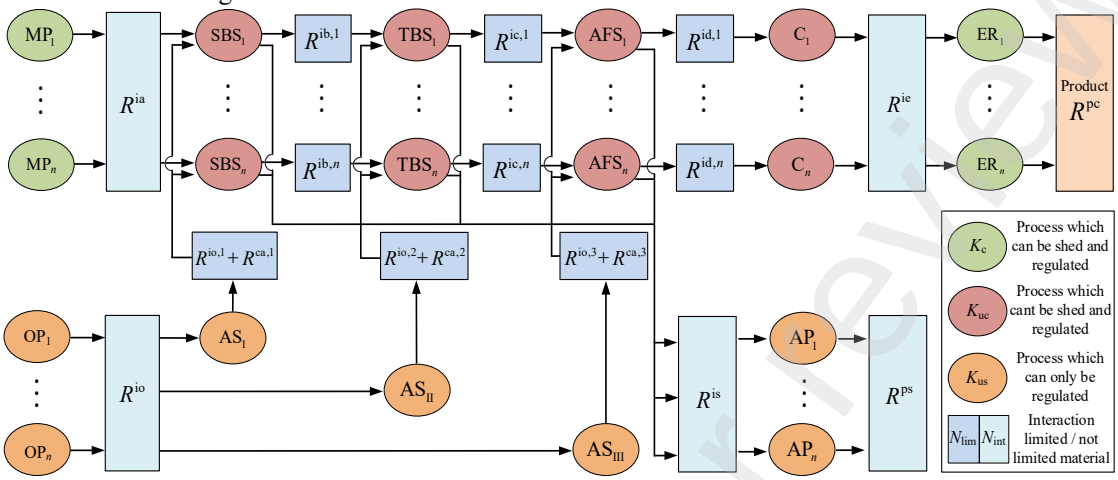


Fig. 1. Conventional RTN model for the ECL. R^{ia}, R^{ib}, R^{ic}, R^{id}, R^{ie}, R^{io}, R^{ca}, R^{is}, R^{ps}, and R^{pc} represent raw materials, blister copper, matte, anode copper, solid anode plates, oxygen, compressed air, sulfur-containing gas, sulfuric acid, and product cathode copper, respectively. The restricted and unconstrained material sets are denoted by N_{lim}={ib,ic,id,ca} and N_{int}={ia,ie,io,is,ps}, respectively.

$$R^{n,t,\omega} = R^{n,t-1,\omega} + \sum_{k\in K}\sum_{i\in I_k} v^{n,k,i}\, y^{k,i,t,\omega} \quad \forall n \in N_{\text{int}}, \omega \tag{10}$$

$$R^{n,t,\omega} = \sum_{i\in I_k} (R^{n,i,t-1,\omega} + \sum_{k\in K} v^{n,k,i}\, y^{k,i,t,\omega}) \;\forall n \in N_{\text{lim}}, \omega \tag{11}$$

$$\underline{R}^{n} \le R^{n,t,\omega} \le \overline{R}^{n} \quad \forall n,t,\omega \tag{12}$$

where $R^{n,t,\omega}$, $v^{n,k,i}$ denote the material inventory and its production or consumption rate. n, N_{int}, N_{lim} represent the material index, the set of materials unconstrained by production lines, and the set of materials restricted to specific lines. $\underline{R}^{n}, \overline{R}^{n}$ denotes the material storage boundaries.

For the ER stage characterized by thermal inertia, the production efficiency of R^{pc} depends on electrolytic power, electrolyte concentration, and temperature [18]. Since continuous replenishment maintains a stable concentration in practice, this paper focuses on the complex "power-temperature-efficiency" coupling. As indicated in [18], electrolytic efficiency approximates a linear function of temperature within a narrow operational band. Consequently, the electrolyte temperature, its boundaries, and temperature-efficiency coupling are formulated as Eqs. (13)-(15):

$$T^{\text{ER},i,t,\omega} = T^{\text{ER},i,t-1,\omega} + \frac{\Delta t}{c^{\text{ER},i} m^{\text{ER},i}} (P^{\text{ER},i,t,\omega} x_{\text{heat}} - Q^{\text{ER},i}) \quad \forall t,\omega \tag{13}$$

$$\underline{T}^{\text{ER},i} \le T^{\text{ER},i,t,\omega} \le \overline{T}^{\text{ER},i} \quad \forall t,\omega \tag{14}$$

$$\eta^{\text{ER},i,t,\omega} = \eta^{\text{ER},i}_{\text{rate}} - e^{\text{ER},i} (T^{\text{ER},i}_{\text{rate}} - T^{\text{ER},i,t,\omega}) \quad \forall t,\omega \tag{15}$$

where $T^{\text{ER},i,t,\omega}$, $T^{\text{ER},i}_{\text{rate}}$, $Q^{\text{ER},i}$, x_{heat}denote the electrolyte temperature, rated temperature, heat dissipation power, and heat conversion efficiency, respectively. Δt represents the 15-minute time interval. $c^{\text{ER},i}$,$m^{\text{ER},i}$ are the specific heat capacity and mass of the electrolyte, with $\underline{T}^{\text{ER},i}$,$\overline{T}^{\text{ER},i}$ indicating its temperature boundaries. Furthermore, $\eta^{\text{ER},i,t,\omega}$,$\eta^{\text{ER},i}_{\text{rate}}$, $e^{\text{ER},i}$ denote the electrolytic efficiency, rated electrolytic efficiency, and

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temperature-efficiency coefficient, respectively.

Thus, the equipment- and temperature-dependent R^{pc} yield and production constraints are formulated as Eqs. (16)-(17):

$$R^{\mathrm{pc},t,\omega} = R^{\mathrm{pc},t-1,\omega} + \sum_{i\in I_{\mathrm{ER}}} v^{\mathrm{pc},\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} \eta^{\mathrm{ER},i,t,\omega} \quad \forall t,\omega \tag{16}$$

$$R^{\mathrm{pc},T,\omega} - R^{\mathrm{pc},0,\omega} \geq R_{\mathrm{aim}} \quad \forall \omega \tag{17}$$

where $R^{\mathrm{pc},T,\omega}$, $R^{\mathrm{pc},0,\omega}$ denote the final and initial R^{pc} inventory of the production cycle, while R_{aim} represents the target yield.

To embed the temperature-dependent production efficiency into the RTN, an auxiliary variable $z^{\mathrm{ER},i,t,\omega}$ is introduced to represent the product $y^{\mathrm{ER},i,t,\omega}\eta^{\mathrm{ER},i,t,\omega}$. Combining the power-temperature dynamics in Eq. (13) with the efficiency relation in Eq. (15) gives Eq. (18):

$$\begin{aligned} z^{\mathrm{ER},i,t,\omega} &= y^{\mathrm{ER},i,t,\omega}\eta^{\mathrm{ER},i,t,\omega} \\ &= \frac{P_{\mathrm{rate}}^{\mathrm{ER},i} e^{\mathrm{ER},i} x_{\mathrm{heat}} \Delta t}{c^{\mathrm{ER},i} m^{\mathrm{ER},i}} (y^{\mathrm{ER},i,t,\omega})^2 + e^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} T^{\mathrm{ER},i,t-1,\omega} + (\eta_{\mathrm{rate}}^{\mathrm{ER},i} - e^{\mathrm{ER},i} T_{\mathrm{rate}}^{\mathrm{ER},i} - \frac{Q^{\mathrm{ER},i,t} e^{\mathrm{ER},i} \Delta t}{c^{\mathrm{ER},i} m^{\mathrm{ER},i}}) y^{\mathrm{ER},i,t,\omega} \\ &= A^{\mathrm{ER},i} (y^{\mathrm{ER},i,t,\omega})^2 + E^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} T^{\mathrm{ER},i,t-1,\omega} + F^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} \end{aligned} \tag{18}$$

where $A^{\mathrm{ER},i}$, $E^{\mathrm{ER},i}$, $F^{\mathrm{ER},i}$ are constant coefficients determined by the production and thermodynamic parameters. The resulting coupling contains a quadratic term and an intertemporal bilinear term, which turns the RTN formulation into a nonconvex MINLP and may hinder reliable solution. A targeted linearization framework is therefore developed to obtain a tractable mixed-integer linear formulation.

2.3 Hybrid SOS2-McCormick Linearization Framework for ThE-RTN Non-convex Couplings

As previously analyzed, R^{pc} efficiency involves quadratic power terms and time-varying bilinearities. Accordingly, a hybrid linearization framework is proposed to ensure both computational tractability and physical accuracy. Specifically, the SOS2 is employed for high-precision linearization of the quadratic terms. For the bilinear terms, McCormick envelopes are utilized for accurate convex relaxation, leveraging the independent boundaries of temperature and power across adjacent intervals. The framework is formulated as Eq. (19).

In the first part, the feasible region of power regulation states is given by Eq. (20). The SOS2 method approximates a nonlinear point as a convex combination of adjacent vertices. Specifically, the non-zero power interval is discretized into N_{p} ordered breakpoints in set $\{\mathbb{Y}_1,\mathbb{Y}_2,...\mathbb{Y}_{N_{\mathrm{p}}}\}$. Evaluating function $f(y^{\mathrm{ER},i})=A^{\mathrm{ER},i}y^{\mathrm{ER},i2}+F^{\mathrm{ER},i}y^{\mathrm{ER},i}$ at these points yields set $\{\mathbb{Z}_1,\mathbb{Z}_2,...\mathbb{Z}_{N_{\mathrm{p}}}\}$, where $\mathbb{Z}_j = f(\mathbb{Y}_j)$. By introducing non-negative weights and strictly restricting at most two adjacent weights to be non-zero, the quadratic terms are accurately approximated. The reconstructed constraints are formulated as Eqs. (21)-(24):

$$z^{\mathrm{ER},i,t,\omega} = \underbrace{[A^{\mathrm{ER},i}(y^{\mathrm{ER},i,t,\omega})^2 + F^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega}]}_{\text{Part 1: SOS2}} + \underbrace{E^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} T^{\mathrm{ER},i,t-1,\omega}}_{\text{Part 2: McCormick}} \tag{19}$$

$$y^{\mathrm{ER},i,t,\omega} \in \{0\} \cup [\underline{y}^{\mathrm{ER},i}, \overline{y}^{\mathrm{ER},i}] \quad \forall t,\omega \tag{20}$$

$$0 \leq \lambda^{i,j,t,\omega} \leq s^{\mathrm{ER},i,t,\omega} \quad \forall j \in \{1,...,N_{\mathrm{p}}\}, t,\omega \tag{21}$$

$$y^{\mathrm{ER},i,t,\omega} = \sum_{j=1}^{N_{\mathrm{p}}} \lambda^{i,j,t,\omega} \cdot \mathbf{Y}^{i,j} \quad \forall t,\omega \tag{22}$$

$$Z_{\mathrm{fix}}^{\mathrm{ER},i,t,\omega} = \sum_{j=1}^{N_{\mathrm{p}}} \lambda^{i,j,t,\omega} \cdot \mathbb{Z}^{i,j,\omega} \quad \forall t,\omega \tag{23}$$

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$$\sum_{j=1}^{N_{\mathrm{p}}} \lambda^{i,j,t,\omega} = s^{\mathrm{ER},i,t,\omega} \quad \{\lambda^{i,1,t,\omega},\ldots,\lambda^{i,N_{\mathrm{p}},t,\omega}\} \in \mathrm{SOS2} \tag{24}$$

where j denotes the ordered breakpoint index, while $\lambda^{i,j,t,\omega}$, $Z_{\mathrm{fix}}^{\mathrm{ER},i,t,\omega}$ represent the weight coefficients and the combined approximation. Eq. (21) bounds the interpolation weights. Eqs. (22) and (23) reconstruct the power-regulation state and the function value, respectively. Eq. (24) links the weight sum to the commitment status and imposes the SOS2 condition, allowing at most two adjacent weights to be non-zero and forcing all weights to zero when the unit is off.

Utilizing McCormick envelopes for the bounded independent bilinearities, the relaxation constraints and the resulting linearized efficiency are given by Eqs. (25)-(26):

$$\begin{cases} W^{\mathrm{ER},i,t,\omega} \geq \underline{T}^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} \\ W^{\mathrm{ER},i,t,\omega} \geq \overline{T}^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} + \overline{y}^{\mathrm{ER},i} T^{\mathrm{ER},i,t-1,\omega} - \overline{T}^{\mathrm{ER},i} \overline{y}^{\mathrm{ER},i} \\ W^{\mathrm{ER},i,t,\omega} \leq \underline{T}^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} + \overline{y}^{\mathrm{ER},i} T^{\mathrm{ER},i,t-1,\omega} - \underline{T}^{\mathrm{ER},i} \overline{y}^{\mathrm{ER},i} \\ W^{\mathrm{ER},i,t,\omega} \leq \overline{T}^{\mathrm{ER},i} y^{\mathrm{ER},i,t,\omega} \end{cases} \quad \forall t,\omega \tag{25}$$

$$z^{\mathrm{ER},i,t,\omega} = Z_{\mathrm{fix}}^{\mathrm{ER},i,t,\omega} + E^{\mathrm{ER},i} \cdot W^{\mathrm{ER},i,t,\omega} \quad \forall t,\omega \tag{26}$$

where $W^{\mathrm{ER},i,t,\omega}$ denotes the auxiliary variable introduced by the McCormick relaxation. Eqs. (20)-(26) form the SOS2-McCormick framework for the 'power-temperature-efficiency' coupling, successfully linearizing R^{pc} efficiency.

The overall error bound $e_{\mathrm{S\text{-}C}}$ of this framework comprises the SOS2 interpolation error e_{S} and the McCormick relaxation error e_{C}, with its theoretical upper limit given by Eq. (27):

$$e_{\mathrm{S\text{-}C}}^{i} \leq \underbrace{\frac{1}{8} h^2 \max |f''(y^{\mathrm{ER},i})|}_{\text{Part 1: } e_{\mathrm{S}}} + \underbrace{\frac{|E^{\mathrm{ER},i}|}{4} \Delta y^{\mathrm{ER},i} \Delta T^{\mathrm{ER},i}}_{\text{Part 2: } e_{\mathrm{C}}} \tag{27}$$

where h denotes the discretization step size, while $\Delta y^{\mathrm{ER},i}$, $\Delta T^{\mathrm{ER},i}$ represent the interval lengths of power bound and temperature. Since the first part is quadratic, the SOS2 error depends on h with an $O(h^2)$ convergence order. Given the narrow operational bounds of both power and temperature, the upper error bound of this framework is tightly constrained.

Collecting the operating, material-state, thermodynamic state, and auxiliary variables introduced above, the ThE-RTN physical scheduling decision vector $\boldsymbol{X}_{\mathrm{phy}}$ is defined by Eq. (28), where the bold symbols collect the corresponding decision variables over all applicable indices.

$$\boldsymbol{X}_{\mathrm{phy}} = \mathrm{col}\{\mathbf{x}, \mathbf{y}, \mathbf{s}, \Delta\mathbf{y}, \mathbf{u}, \mathbf{d}, \mathbf{R}, \mathbf{T}, \boldsymbol{\eta}, \mathbf{z}, \boldsymbol{\lambda}^{\mathrm{SOS2}}, \mathbf{Z}, \mathbf{W}\} \tag{28}$$

Accordingly, the proposed ThE-RTN combines plant-wide inter-stage material constraints with the ER-specific power-temperature-efficiency dynamics. The former preserves material-network feasibility, whereas the latter captures the cross-period effects of power regulation on temperature and production yield. The hybrid SOS2-McCormick formulation provides an MILP-representable approximation whose error is explicitly bounded by Eq. (27). The variable vector $\boldsymbol{X}_{\mathrm{phy}}$, together with Eqs. (1)-(26), defines the physical scheduling feasible domain used by the subsequent SC-IGDT model. The conventional RTN is extensible across process industries through reconfiguration of task-resource relationships. Since the proposed ThE-RTN retains this network structure and augments it with process-specific thermodynamic couplings, it preserves the structural extensibility of RTN; application to another thermal-inertia PIL requires reconfiguration of the RTN topology and identification of the corresponding thermodynamic equations and parameters. The resulting ThE-RTN framework is illustrated in Fig. 2.

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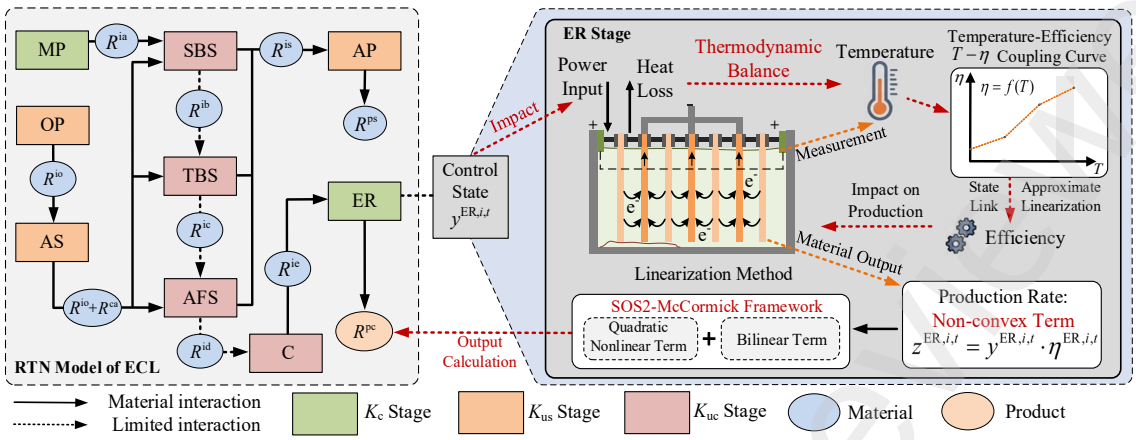


Fig. 2. ThE-RTN model and algorithmic topology for PILs.

3. SC-IGDT Co-optimization of Production Scheduling and Bidding under PV-Price Uncertainties

Building on the ThE-RTN, this section develops an SC-IGDT model for coordinated production scheduling and bidding for ECL participation in the DA-RT spot market under PV and price uncertainties, considering deviation settlement and arbitrage-recovery rules. Scenario-based SP-CVaR and IGDT are used to address PV and price uncertainties, respectively. A joint-vertex affine-epigraph reformulation and outer bisection are employed to handle the SIP and bilinear terms, respectively.

The model assumes that: 1) the ECL is a price taker and does not affect market-clearing prices; 2) PV uncertainty is represented by historical scenarios, whereas DA-RT price uncertainty is described by an IGDT uncertainty set; and 3) the ECL submits volume-only DA bids, while existing mid-to-long-term (MLT) contract quantities and prices are fixed and used only for deviation settlement.

3.1 DA-RT Settlement and PV-price Uncertainty Modeling

Spot settlement for PILs spans three temporal stages: MLT, DA, and RT. Specifically, MLT contracts lock in baseline volumes and prices. DA bidding schedules next-day consumption, with deviations from MLT volumes settled at DA prices. Finally, RT operations execute the schedules, clearing actual consumption deviations from DA bids at RT prices. The multi-stage costs are formulated as Eqs. (29)-(33):

$$C^{h}_{\mathrm{MLT}} = C^{h}_{\mathrm{BAS}}\alpha_{\mathrm{MLT}} \quad \forall h \tag{29}$$

$$F^{h}_{\mathrm{MLT}} = C^{h}_{\mathrm{MLT}}\lambda^{h}_{\mathrm{MLT}} \quad \forall h \tag{30}$$

$$F^{h}_{\mathrm{DA,Dev}} = (C^{h}_{\mathrm{BID}} - C^{h}_{\mathrm{MLT}})\lambda^{h}_{\mathrm{DA}} \quad \forall h \tag{31}$$

$$F^{h,\omega}_{\mathrm{RT,Dev}} = (C^{h,\omega}_{\mathrm{ACT}} - C^{h}_{\mathrm{BID}})\lambda^{h}_{\mathrm{RT}} \quad \forall h,\omega \tag{32}$$

$$C^{h,\omega}_{\mathrm{ACT}} = \sum_{t\in T_h}\Big(\sum_{k\in K}\sum_{i\in I_k} P^{k,i,t,\omega} - P^{t,\omega}_{\mathrm{PV}}\Big)\Delta t \quad \forall h,\omega \tag{33}$$

where C^{h}_{MLT}, C^{h}_{BID}, $C^{h,\omega}_{\mathrm{ACT}}$ denote the volumes for MLT contracts, DA bids, and RT actual consumption, respectively; C^{h}_{BAS}, α_{MLT} represent the rated power and MLT volume ratio. $\lambda^{h}_{\mathrm{MLT}}$, $\lambda^{h}_{\mathrm{DA}}$, $\lambda^{h}_{\mathrm{RT}}$ denote the prices for the MLT, DA, and RT stages. F^{h}_{MLT}, $F^{h}_{\mathrm{DA,Dev}}$, $F^{h,\omega}_{\mathrm{RT,Dev}}$ are the costs for the MLT and DA/RT deviation settlements, while $P^{t,\omega}_{\mathrm{PV}}$ is the PV output. Eqs. (29)-(30) formulate the MLT baseline lock-in. Eqs. (31)-(32) define the DA and RT deviation settlements, and Eq. (33) reflects the PV-induced uncertainty in actual RT consumption.

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Furthermore, spot markets impose deviation limits and recover settlement gains arising from excessive deviations [23]. PILs face asymmetric arbitrage-recovery penalties if RT consumption exceeds the penalty-free deadband of DA bids. Specifically, this encompasses over-bidding (over-reporting DA volumes during high RT prices) and under-bidding (under-reporting DA volumes during low RT prices) arbitrages. Accordingly, the penalty formulations are given by Eqs. (34)-(36):

$$\Delta\lambda_{\mathrm{ob}}^{h} = \max(0, \lambda_{\mathrm{RT}}^{h} - \lambda_{\mathrm{DA}}^{h}) \quad \forall h, \omega \tag{34}$$

$$\Delta\lambda_{\mathrm{ub}}^{h} = \max(0, \lambda_{\mathrm{DA}}^{h} - \lambda_{\mathrm{RT}}^{h}) \quad \forall h, \omega \tag{35}$$

$$\begin{aligned} F_{\mathrm{PEN}}^{h,\omega} = \mu_{\mathrm{PEN}}[&\max(0, C_{\mathrm{BID}}^{h} - (1+V_{\mathrm{PEN}})C_{\mathrm{ACT}}^{h,\omega}) \cdot \Delta\lambda_{\mathrm{ob}}^{h} \\ &+ \max(0, (1-V_{\mathrm{PEN}})C_{\mathrm{ACT}}^{h,\omega} - C_{\mathrm{BID}}^{h}) \cdot \Delta\lambda_{\mathrm{ub}}^{h}] \end{aligned} \quad \forall h, \omega \tag{36}$$

where $\Delta\lambda_{\mathrm{ob}}^{h}$, $\Delta\lambda_{\mathrm{ub}}^{h}$ denotes the price spreads for over/under-bidding, $F_{\mathrm{PEN}}^{h,\omega}$ is the arbitrage penalty, while μ_{PEN}, V_{PEN} represent the recovery and penalty-free ratios, respectively.

To characterize the PV and price uncertainties involved in Eqs. (33)-(36), scenario-based stochastic modeling and IGDT are adopted for PV output and DA-RT prices, respectively. For PV modeling, Monte Carlo simulation first generates raw trajectories from historical forecast errors around a data-driven baseline forecast. K-medoids clustering is then employed for scenario reduction, with actual sampled trajectories retained as medoids to preserve temporal dependence and physical consistency. The resulting typical PV scenarios and their probabilities are formulated in Eq. (37).

$$\Omega = \left\{ \left(\mathbf{P}_{\mathrm{PV}}^{\omega}, \pi^{\omega} \right) \middle| \sum_{\omega \in \Omega} \pi^{\omega} = 1, \pi^{\omega} \in (0,1] \right\} \tag{37}$$

where Ω denotes the typical PV scenario set, while $\mathbf{P}_{\mathrm{PV}}^{\omega}$, π^{ω} represent the PV profile and its probability under scenario ω. Accordingly, C_{BID}^{h} is a first-stage decision shared by all PV scenarios, whereas the scenario-dependent production schedule and $C_{\mathrm{ACT}}^{h,\omega}$ are second-stage recourse decisions.

For spot-price uncertainty, an IGDT uncertainty radius α is introduced around the baseline DA-RT price forecasts. The radius defines the admissible price-deviation set and provides the basis for evaluating the maximum price uncertainty that can be tolerated under a prescribed cost requirement. Thus, the DA-RT price uncertainty set is formulated in Eq. (38).

$$U(\alpha) = \left\{ \tilde{\lambda}^{h} \middle| \begin{array}{c} (1-\alpha)\lambda_{\mathrm{DA}}^{h} \le \tilde{\lambda}_{\mathrm{DA}}^{h} \le (1+\alpha)\lambda_{\mathrm{DA}}^{h} \\ (1-k_{\mathrm{RT}}\alpha)\lambda_{\mathrm{RT}}^{h} \le \tilde{\lambda}_{\mathrm{RT}}^{h} \le (1+k_{\mathrm{RT}}\alpha)\lambda_{\mathrm{RT}}^{h} \end{array}, \forall h \right\} \tag{38}$$

where $U(\alpha)$ denotes the DA-RT spot price sets, while $\tilde{\lambda}_{\mathrm{DA}}^{h}$, $\tilde{\lambda}_{\mathrm{RT}}^{h}$ represent the fluctuating prices under α. The coefficient k_{RT} indicates that RT volatility exceeds that of DA. Combining the settlement model and dual-uncertainty representations in Eqs. (29)-(38), the next subsection formulates the SC-IGDT co-optimization model for production scheduling and bidding.

3.2 SC-IGDT Co-Optimization Model for Production Scheduling and Bidding

Based on the above settlement mechanism and PV-price uncertainty modeling, an SC-IGDT co-optimization model for production scheduling and bidding is formulated. Its decision variables are collected as Eq. (39):

$$\boldsymbol{X} = \mathrm{col}\{\boldsymbol{X}_{\mathrm{phy}}, \mathbf{C}_{\mathrm{BID}}, \mathbf{C}_{\mathrm{ACT}}\} \tag{39}$$

where $\mathbf{C}_{\mathrm{BID}}$ and $\mathbf{C}_{\mathrm{ACT}}$ denote the DA bidding volumes and actual electricity consumption under different PV scenarios.

For scenario ω, the worst-case spot operating cost over the price uncertainty set $U(\alpha)$ is defined in Eq. (40):

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$$F^{\omega}_{\text{worst}}(\boldsymbol{X},\alpha)=\max_{\tilde{\boldsymbol{\lambda}}\in U(\alpha)} F^{\omega}_{\text{spot}}(\boldsymbol{X},\tilde{\boldsymbol{\lambda}}) \quad \forall\omega \tag{40}$$

where F^{ω}_{spot} consists of the fixed MLT cost F^{h}_{MLT}, the spot exposure costs $F^{h}_{\text{DA,Dev}}$, $F^{h,\omega}_{\text{RT,Dev}}$, $F^{h,\omega}_{\text{PEN}}$, and the power-regulation costs $F^{h,\omega}_{\text{cpr}}$, $F^{h,\omega}_{\text{ud}}$. Accordingly, the SP-CVaR comprehensive risk cost J_{worst} is formulated as Eqs. (41)-(43):

$$J_{\text{worst}}=(1-\beta)\sum_{\omega\in\Omega}\pi^{\omega}F^{\omega}_{\text{worst}}+\beta\text{CVaR}_{\gamma}(F_{\text{worst}}) \tag{41}$$

$$\text{CVaR}_{\gamma}(F_{\text{worst}})=\zeta+\frac{1}{1-\gamma}\sum_{\omega\in\Omega}z^{\omega}\pi^{\omega} \quad \forall\omega \tag{42}$$

$$z^{\omega}\ge F^{\omega}_{\text{worst}}-\zeta, z^{\omega}\ge 0 \quad \forall\omega \tag{43}$$

where β and γ denote the risk-aversion coefficient and CVaR confidence level. $\text{CVaR}_{\gamma}(F_{\text{worst}})$,$\zeta$, z^{ω} denote the CVaR, VaR, and excess tail-loss variable, respectively. The resulting cost serves as the IGDT performance measure, jointly accounting for PV scenario probabilities, tail risk, and price uncertainty.

Since the MLT contract cost remains fixed over the decision horizon and does not affect bidding or production scheduling, the optimal spot exposure cost at α=0, excluding the MLT cost, is adopted as the nominal benchmark in Eq. (44):

$$J_0=\min_{\boldsymbol{X},\zeta,z}[J_{\text{worst}}|_{\alpha=0}-\sum_{h\in H}F^{h}_{\text{MLT}}] \tag{44}$$

Given an acceptable cost increase δ, the SC-IGDT model maximizes the price uncertainty radius while restricting the spot exposure cost to $(1+\delta)J_0$, which is shown in Eq. (45):

$$\begin{aligned}&\max_{\boldsymbol{X},\alpha,\zeta,z}\quad \alpha\\ &s.t.\quad J_{\text{worst}}-\sum_{h\in H}F^{h}_{\text{MLT}}\le(1+\delta)J_0,\\ &0\le\alpha\le\alpha_{\max}, \text{Eqs.}(1)-(26),(28)-(44)\end{aligned} \tag{45}$$

where $\alpha_{\max}$ denotes the upper bound of the uncertainty-radius search interval. PV uncertainty is characterized by scenario probabilities and CVaR, whereas price uncertainty is represented by the IGDT radius. However, the continuous uncertainty set $U(\alpha)$ introduces infinitely many price-indexed constraints, resulting in a semi-infinite programming problem. Moreover, directly optimizing α generates bilinear terms involving C^{h}_{BID}, $C^{h,\omega}_{\text{ACT}}$. The next subsection therefore employs a joint-vertex affine-epigraph reformulation and an outer radius search to determine the optimal uncertainty radius α^{*}.

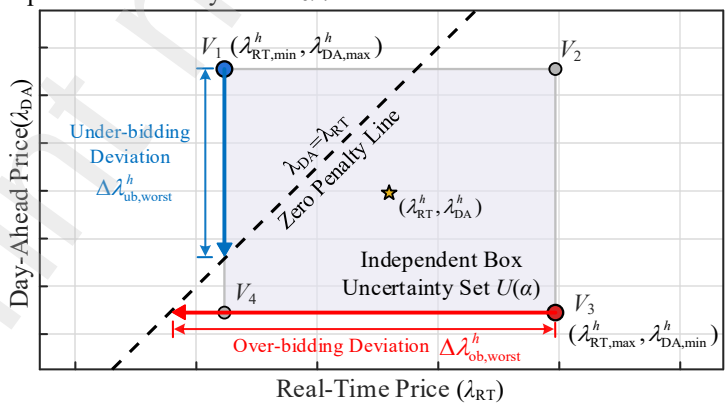


Fig. 3. Joint vertices and worst-case deviation spreads of the box-shaped DA-RT price uncertainty set.

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3.3 Joint-Vertex Affine-Epigraph Reformulation and Outer Bisection Algorithm

To address the SIP problem, a fixed price-uncertainty radius α is first considered. As indicated by Eqs. (31)-(32), the DA and RT deviation settlement costs are affine functions of prices, while the deviation-arbitrage penalty in Eq. (36) is the maximum of several affine functions. Therefore, for given bidding and scheduling decisions, the hourly spot settlement cost is convex and piecewise affine in the DA-RT prices.

As shown in Fig. 3, the DA-RT price intervals form a two-dimensional box uncertainty set, whose vertex set $V^h(\alpha) = \{V1, V2, V3, V4\}$ represents the combinations of the DA and RT price bounds. The dashed line denotes the zero-spread boundary. According to Eq. (36), the worst-case penalty search identifies the maximum positive deviation of a price point from the zero-spread line within this set, which is attained at a vertex. The red and blue segments represent the worst-case spreads under over-bidding and under-bidding.

The deviation settlement cost at vertex v is defined as Eq. (46). An epigraph variable $\theta^{h,\omega}$ is introduced to represent the maximum deviation settlement and arbitrage-recovery cost over the joint price vertices, shown in Eq. (47). Accordingly, the worst-case spot operating cost in Eq. (40) is reformulated as Eq. (48).

$$L^{h,\omega,v} = F^{h,v}_{\mathrm{DA,Dev}} + F^{h,\omega,v}_{\mathrm{RT,Dev}} \quad \forall v \in V^h(\alpha) \tag{46}$$

$$\begin{cases} \theta^{h,\omega} \geq L^{h,\omega,v} + \mu_{\mathrm{PEN}} \Delta\lambda^{h,v}_{\mathrm{ub}} [(1 - V_{\mathrm{PEN}}) C^{h,\omega}_{\mathrm{ACT}} - C^{h}_{\mathrm{BID}}] \\ \theta^{h,\omega} \geq L^{h,\omega,v} + \mu_{\mathrm{PEN}} \Delta\lambda^{h,v}_{\mathrm{ob}} [C^{h}_{\mathrm{BID}} - (1 + V_{\mathrm{PEN}}) C^{h,\omega}_{\mathrm{ACT}}] \quad \forall v \in V^h(\alpha) \\ \theta^{h,\omega} \geq L^{h,\omega,v} \end{cases} \tag{47}$$

$$F^{\omega}_{\mathrm{worst}} = \sum_{h \in H} (F^{h}_{\mathrm{MLT}} + F^{h,\omega}_{\mathrm{cpr}} + F^{h,\omega}_{\mathrm{ud}} + \theta^{h,\omega}) \quad \forall \omega \tag{48}$$

For fixed α, all vertex prices and corresponding price spreads are constants, and the above epigraph constraints are linear. Hence, the continuous worst-case price search is replaced by a finite set of joint-vertex constraints, converting the fixed-radius SIP into an equivalent MILP. The extended decision-variable vector, including the original scheduling and auxiliary variables, is defined as Eq. (49):

$$\tilde{\boldsymbol{X}} = \mathrm{col}\{\boldsymbol{X}, \boldsymbol{\theta}, \mathbf{F}_{\mathrm{worst}}, \mathbf{z}^{\mathrm{CVaR}}, \boldsymbol{\zeta}\} \tag{49}$$

For a given radius α, the optimal value of the fixed-radius affine-epigraph MILP is defined as Eq. (50):

$$\Phi(\alpha) = \min_{\tilde{\boldsymbol{X}}} [J_{\mathrm{worst}}(\tilde{\boldsymbol{X}}; \alpha) - \sum_{h \in H} F^{h}_{\mathrm{MLT}}] \tag{50}$$

where $\Phi(\alpha)$ denotes the minimum worst-case spot-exposure risk cost attainable through optimal bidding and scheduling under radius α. If any decision satisfies the IGDT constraint, $\Phi(\alpha)$ cannot exceed its corresponding cost; conversely, the optimizer associated with $\Phi(\alpha)$ provides a feasible solution to the original problem. Therefore, the conventional SC-IGDT problem is equivalently expressed as Eq. (51):

$$\alpha^{*} = \max\{\alpha \in [0, \alpha_{\max}] : \Phi(\alpha) \leq (1 + \delta) J_0\} \tag{51}$$

Furthermore, increasing α expands the price uncertainty set; hence, the worst-case cost under any fixed decision cannot decrease. Algorithm 1 therefore applies outer bisection to determine the maximum feasible IGDT radius α^*.

Integrating the ThE-RTN with SC-IGDT, the framework jointly determines DA bids, production schedules, and the maximum feasible price-uncertainty radius, while joint-vertex affine-epigraph reformulation and outer bisection ensure tractability. The framework is illustrated in Fig. 4.

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Algorithm 1: Outer Bisection for the Maximum Feasible IGDT Radius

Input: Cost tolerance δ, search bound $\alpha_{\max}$, and radius tolerance ε_α
Output: Maximum feasible radius α^*
Set $J_0 \leftarrow \Phi(0)$, $\alpha_L \leftarrow 0$, $\alpha_U \leftarrow \alpha_{\max}$.
if $\Phi(\alpha_U) \le (1+\delta)J_0$ then
  return $\alpha^* \leftarrow \alpha_U$.
end if
while $\alpha_U - \alpha_L > \varepsilon_\alpha$ do
  Set $\bar{\alpha} \leftarrow (\alpha_L + \alpha_U)/2$.
  Solve the fixed-$\bar{\alpha}$ joint-vertex affine-epigraph MILP to obtain $\Phi(\bar{\alpha})$.
  if $\Phi(\bar{\alpha}) \le (1+\delta)J_0$ then
    Set $\alpha_L \leftarrow \bar{\alpha}$.
  else
    Set $\alpha_U \leftarrow \bar{\alpha}$.
  end if
end while
return $\alpha^* \leftarrow \alpha_L$.

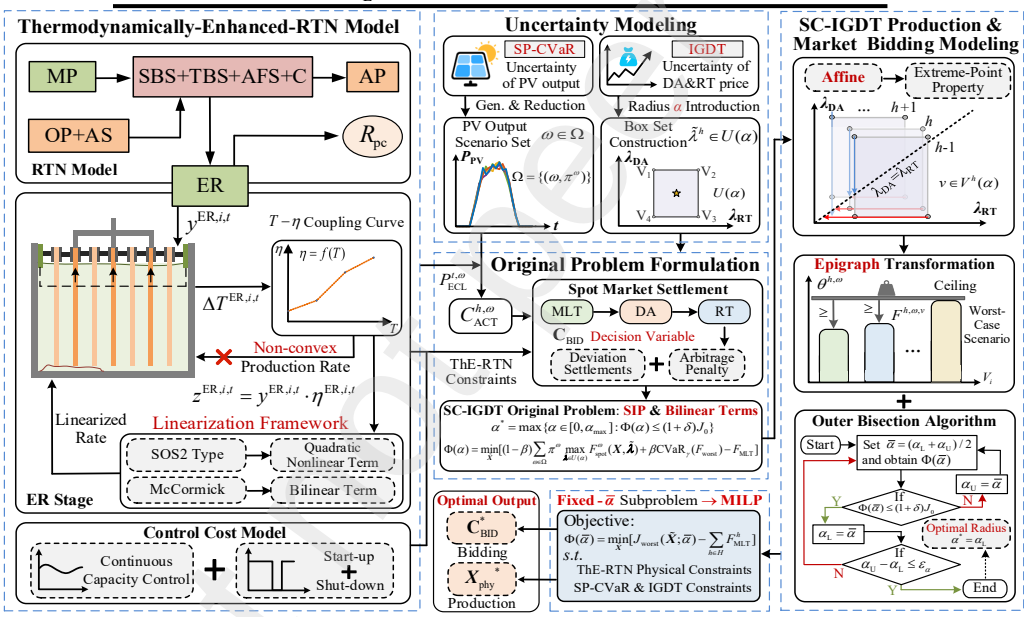


Fig. 4. Schematic of the coordinated production-scheduling and bidding framework.

4. Case Study

To validate the proposed coordinated production-scheduling and bidding framework, a case study based on an actual ECL in Yunnan, China, is conducted. The model is formulated in MATLAB R2023a via YALMIP and solved by Gurobi 13.0 (MIP gap: 0.01%) on an AMD Ryzen 9 9955HX CPU with 32 GB RAM. The 24-hour production cycle is discretized into 96 periods with a 15-minute time step.

4.1 Parameter Settings

The ECL features a 22 MW rated capacity and a 3 MW PV installation. Except for the ER and AS processes (comprising three series each), all others operate singularly; detailed parameters are listed in Table

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1. Initially, inventories for materials ia, ie, and io are 1000, 10, and 2400 (t or m³), with others initialized to zero. Material boundary constraints refer to [13], with a 24-h production target of 7690 t. The electrolyte temperature is constrained within 65-70°C. The three ER series share identical derived thermodynamic coefficients, with A^{ER}=0.02711125, E^{ER}=0.01, F^{ER}=0.26288875, calculated from the thermodynamic parameters in Eq. (18). The SOS2 approximation uses 3 uniformly spaced breakpoints over [0.75, 1.05], with h=0.15.

Figs. 5 and 6 depict the ECL load baseline, the PV scenario set for the trading day, and the MLT and forecasted DA/RT baseline prices. Market and risk parameters are configured as follows: the MLT volume ratio α_{MLT} is 0.85; the arbitrage recovery μ_{PEN} and penalty-free ratios V_{PEN} are 0.8 and 0.1, respectively; the RT volatility coefficient k_{RT} is 1.2; and the CVaR weight β and confidence level γ are 0.2 and 0.9.

Table 1. Operational parameters of ECL processes.

Stage	Power (kW)	Boundary	Control Cost (CNY)			
			$\eta_{up}^{k,i}$	$\eta_{down}^{k,i}$	$\eta_{su}^{k,i}$	$\eta_{sd}^{k,i}$
MP	1579.3	[0.85,1.05]	0.021	0.01	10	20
SBS	1643.7	/	/	/	/	/
TBS	1643.7	/	/	/	/	/
AFS	1029.6	/	/	/	/	/
C	1110	/	/	/	/	/
ER_1	2168.9	[0.75,1.05]	0.03	0.014	20	40
ER_2	2168.9	[0.75,1.05]	0.03	0.014	20	40
ER_3	2168.9	[0.75,1.05]	0.03	0.014	20	40
OP	4556	[0.8,1.05]	0.021	0.011	/	/
AS_1	535.6	[0.95,1]	0.003	0.003	/	/
AS_2	465.6	[0.9,1]	0.003	0.003	/	/
AS_3	345.6	[0.8,1]	0.003	0.003	/	/
AP	2590.2	[0.9,1.05]	0.036	0.018	/	/

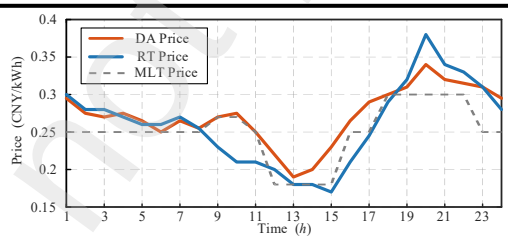


Fig. 5. MLT, DA, and RT price curves.

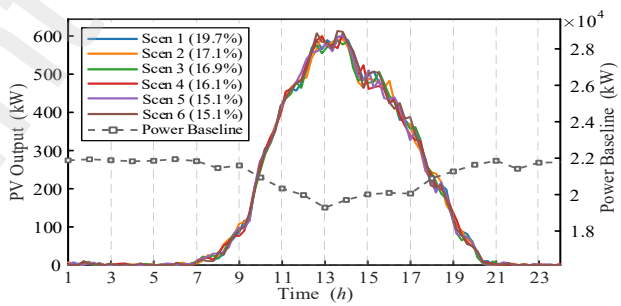


Fig. 6. ECL load baseline and typical PV output scenarios.

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*4.2 Determination and Interpretation of the Optimal IGDT Radius α**

With the cost tolerance set to δ=0.1, the nominal spot-exposure cost is J_0=16,475.20 CNY, yielding a cost limit of 18,122.72 CNY. As reported in Table 2, the final bisection stage converges after eight iterations and obtains α^*=3.781% within 148.9 s. At this radius, $\Phi(\alpha^*)$=18,122.15 CNY satisfies the prescribed cost limit, whereas the adjacent upper bound α_U=3.789% gives $\Phi(\alpha_U)$=18,124.68 CNY and becomes infeasible. Therefore, α^* is the maximum feasible price-uncertainty radius within the specified bisection tolerance.

Table 2. Determination of the maximum feasible IGDT radius.

Metric	Value
α_{max} (%)	50
ε_α	1×10^{-4}
J_0 (CNY)	16475.20
$(1+\delta)J_0$ (CNY)	18122.72
α^* (%)	3.781
$\Phi(\alpha^*)$ (CNY)	18122.15
α_U (%)	3.789
$\Phi(\alpha_U)$ (CNY)	18124.68

Fig. 7 further illustrates the price uncertainty bands and corresponding bidding decision. The bid generally remains near the expected net consumption when deviation exposure dominates. In contrast, substantial downward bid adjustments, including zero bids, arise when favorable DA-RT price spreads persist throughout the uncertainty range. This contrast reflects the trade-off between deviation exposure and persistent settlement incentives under the prescribed cost budget. The zero-bidding decisions are therefore examined through their marginal settlement costs. For the zero-bidding periods, the lower DA-price bound remains above the upper RT-price bound under α^*. Within the penalty-free region, the marginal settlement cost equals $\lambda_{DA}^{h,v} - \lambda_{RT}^{h,v}$. Once the bid falls below the penalty-free lower bound, it becomes Eq. (52).

$$\nabla F = \partial F^{h,\omega,v} / \partial C_{BID}^{h} = (1-\mu_{PEN})(\lambda_{DA}^{h,v} - \lambda_{RT}^{h,v}) \tag{52}$$

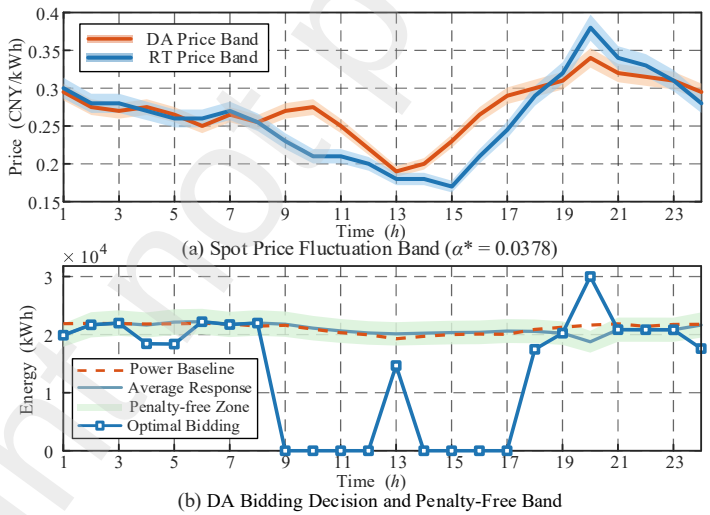


Fig. 7. Price uncertainty bands and the corresponding bidding decision.

This expression applies when $C_{BID}^{h} < (1 - V_{PEN})C_{ACT}^{h,\omega}$; crossing C_{MLT}^{h} only changes the ECL from a DA buyer to a seller. Since gradients remain positive at all joint price vertices, reducing the DA bid decreases the worst-case settlement cost despite the arbitrage-recovery charge. For example, at h = 10, the DA-price lower bound still exceeds the RT-price upper bound by 0.0451 CNY/kWh under α^*. Hence, the optimal bid reaches its lower bound of zero. Conversely, at h=20, the RT-price lower bound exceeds the DA-price upper bound

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under α^*. The marginal cost remains negative at all joint vertices, so increasing the DA bid reduces the worst-case settlement cost despite arbitrage recovery, driving the bid beyond penalty-free band. Together, these decisions show that SC-IGDT does not impose uniformly conservative bidding; instead, it adjusts the bidding direction according to persistent DA-RT price spreads, thereby preserving favorable settlement opportunities while satisfying the prescribed cost budget.

To interpret the cost-tolerance parameter, four values of δ are considered, with the resulting maximum tolerable price deviations reported in Table 3. Increasing δ from 0.04 to 0.15 raises α^* from 1.188% to 6.516%. Their nonlinear relationship reflects the combined effects of production constraints and piecewise settlement rules, providing ECL operators with a direct reference for selecting the price-risk protection level according to their acceptable cost budget.

Table 3. Maximum tolerable price deviations.

δ	$(1+\delta)\,J_0$ (CNY)	α^* (%)
0.04	17134.21	1.188
0.08	17793.22	2.781
0.1	18122.72	3.781
0.15	18946.48	6.516

4.3 Effectiveness of Uncertainty Management

To evaluate price-uncertainty treatments, Table 4 compares four methods under identical ThE-RTN constraints, PV scenarios, CVaR settings, and settlement rules, with a budget exceedance uniformly defined as an OOS spot-exposure cost above $(1+\delta)J_0$. SC-RO uses α_{RO}=12.263% from the 95th percentile of daily maximum normalized training errors; SC-DRO uses eight representative error trajectories and a 95%-confidence probability box with ε_q=0.13426; and SC-IGDT uses δ=0.1, yielding α^*=3.781%. SC-RO and SC-DRO are calibrated on 160 synthetic trajectories, and all methods are evaluated on 5000 independent OOS trajectories.

Table 4. Out-of-sample performance of different price-uncertainty treatments.

Method	Nominal SC-CVaR	SC-RO	SC-DRO	SC-IGDT
Nominal cost (CNY)	16475	17789	16499	16911
Nominal cost premium (%)	0	7.98	0.15	2.65
OOS mean (CNY)	16596	17789	16534	16914
OOS $CVaR_{0.95}$ (CNY)	17263	17988	17017	17189
OOS P99 (CNY)	17407	18008	17140	17224
OOS maximum (CNY)	19114	18164	17618	17593
Budget exceedances (/5000)	2	3	0	0

At the original price-error scale, Table 4 shows that SC-DRO achieves the lowest OOS mean and $CVaR_{0.95}$. SC-RO incurs the largest nominal cost premium of 7.98%, reflecting the conservatism of its calibrated uncertainty set, whereas SC-IGDT requires a moderate premium of 2.65%. Compared with Nominal SC-CVaR, SC-IGDT reduces the observed maximum cost by 7.96% and, like SC-DRO, records no budget exceedances among 5000 OOS trajectories. These results indicate that SC-IGDT improves protection against sampled extreme costs at a moderate nominal cost premium.

To assess sensitivity to larger price deviations, the OOS price-error scale factor is increased from 1.0 to 2.0. Fig. 8 reports the increases in $CVaR_{0.95}$ and P99 relative to each method's results at scale 1.0, while Table 5 summarizes performance under doubled errors. The optimized decisions and calibration parameters remain fixed, and all methods are evaluated using the same 5000 OOS trajectories and budget threshold.

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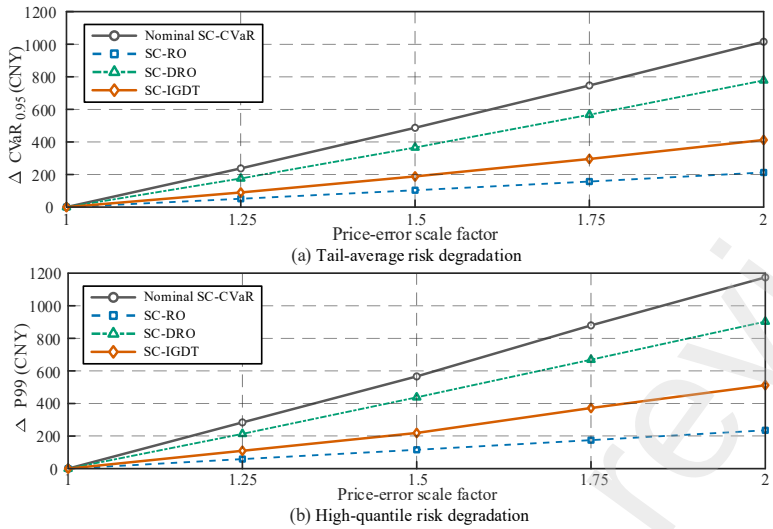


Fig. 8. Increases in out-of-sample tail-risk measures with increasing price-error magnitude: (a)$CVaR_{0.95}$; (b)P99.

Table 5. Out-of-sample performance under doubled price-error magnitude.

Method	Nominal SC-CVaR	SC-RO	SC-DRO	SC-IGDT
OOS mean (CNY)	16837	17792	16667	16931
Mean increase (CNY)	240.4	3.1	133.5	17.8
OOS $CVaR_{0.95}$ (CNY)	18278	18201	17794	17601
OOS P99 (CNY)	18581	18244	18043	17737
OOS maximum (CNY)	21922	18962	19031	18722
Budget exceedances (/5000)	132	229	41	16

As the error scale increases, SC-RO exhibits the smallest increases in $CVaR_{0.95}$ and P99, while SC-IGDT shows smaller increases than Nominal SC-CVaR and SC-DRO. Under doubled errors, SC-IGDT achieves the lowest absolute $CVaR_{0.95}$, P99, and observed maximum cost, with 16 budget exceedances among 5000 trajectories, compared with 41 for SC-DRO. Although SC-DRO retains a lower mean cost, SC-IGDT provides stronger tail-risk protection in this stress test at a lower nominal cost premium than SC-RO. Moreover, IGDT determines the maximum admissible price-deviation radius under a prescribed cost tolerance and uncertainty-set structure without specifying a price-error probability distribution. These results support its suitability for balancing operating economy and adverse-price protection in the present scheduling and bidding problem.

For PV uncertainty management, owing to limited PV penetration and scenario dispersion, varying β from 0 to 0.8 changes α^* by less than 10^{-4}, indicating that price uncertainty dominates the present risk profile. Hence, CVaR mainly serves as an auxiliary safeguard against low-probability PV deviations rather than a primary driver of the current bidding decision.

4.4 Operational Strategy Comparison and Economic Performance

To quantify the incremental economic contributions of the ThE-RTN and SC-IGDT, Table 6 compares three operational strategies under the same case settings and uncertainty radius α^*. Strategy 1 employs self-scheduled conventional RTN with conventional bidding; Strategy 2 employs price-responsive ThE-RTN with conventional bidding; and Strategy 3 adopts the coordinated production-scheduling and bidding framework. Since the MLT cost is fixed across strategies, only the spot-exposure cost components and risk cost are reported. A negative DA deviation cost denotes a settlement credit relative to the MLT position.

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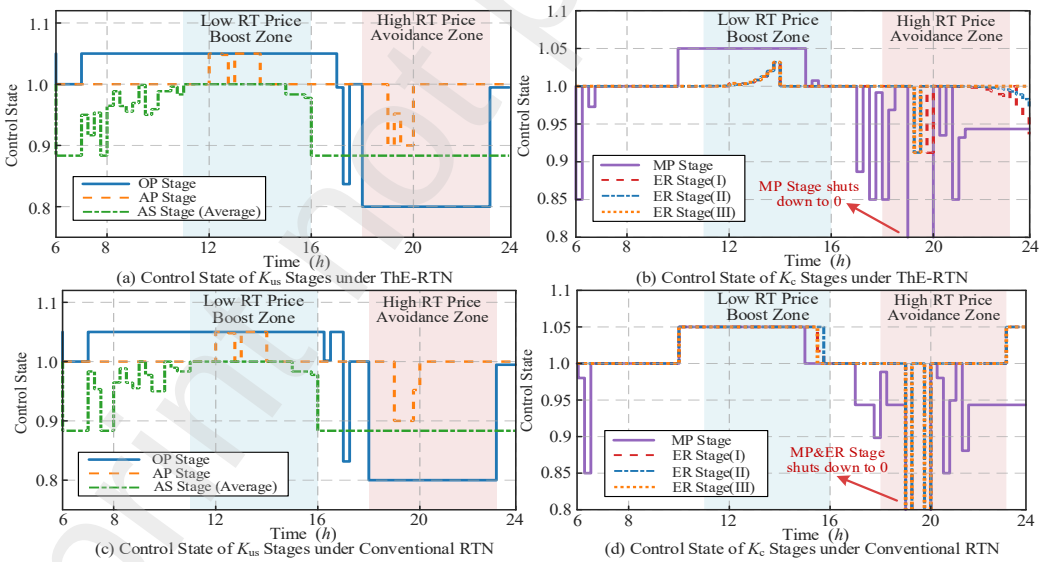
Table 6. Cost decomposition and spot-exposure risk cost of three operating strategies.

Metric	Strategy 1	Strategy 2	Strategy 3
Expected $F_{\mathrm{DA,Dev}}$ (CNY)	22011	19704	-21925
Expected $F_{\mathrm{RT,Dev}}$ (CNY)	2.86	3.12	36843
Expected F_{PEN} (CNY)	0	0	3000.2
Expected $F_{\mathrm{cpr}}+F_{\mathrm{ud}}$ (CNY)	19.33	168.19	197.56
J_{worst}-$\sum F_{\mathrm{MLT}}$ (CNY)	22039	19877	18122
Cost Reduction Rate (%)	/	9.81	17.77

As shown in Table 6, Strategy 2 reduces risk cost from 22,039 to 19,877 CNY, corresponding to a 9.81% decrease relative to Strategy 1. Although the internal-control cost increases from 19.33 to 168.19 CNY, the expected DA deviation cost decreases by 2307 CNY. This indicates that ThE-RTN-based price-responsive scheduling exploits thermodynamically feasible flexibility to reduce market exposure at a limited control cost. Strategy 3 further reduces risk cost to 18,122 CNY, providing an additional saving of 8.83%. Although RT deviation and arbitrage-recovery costs increase, they are more than offset by the DA settlement credit. Hence, ThE-RTN-based scheduling already realizes part of the economic value of physical flexibility, while coordinating production with DA bidding and DA-RT settlement decisions further improves this value. Overall, the proposed strategy reduces risk cost by 17.77% relative to Strategy 1, demonstrating that the ThE-RTN enables executable price-responsive flexibility, while SC-IGDT further realizes its economic value through coordinated production scheduling and bidding.

4.5 Physical Feasibility and Operational Safety of the ThE-RTN

To verify the physical feasibility and operational safety of the ThE-RTN, Figs. 9-11 demonstrate its coordinated power regulation, material and production balance, and temperature-limit compliance under the most probable PV scenario (ω=1), with the conventional RTN schedule thermodynamically replayed for comparison.



(a) Control State of K_{us} Stages under ThE-RTN
(b) Control State of K_{c} Stages under ThE-RTN
(c) Control State of K_{us} Stages under Conventional RTN
(d) Control State of K_{c} Stages under Conventional RTN

Fig. 9. Power-control states of the main ECL stages under (a)-(b) the ThE-RTN and (c)-(d) the conventional RTN.

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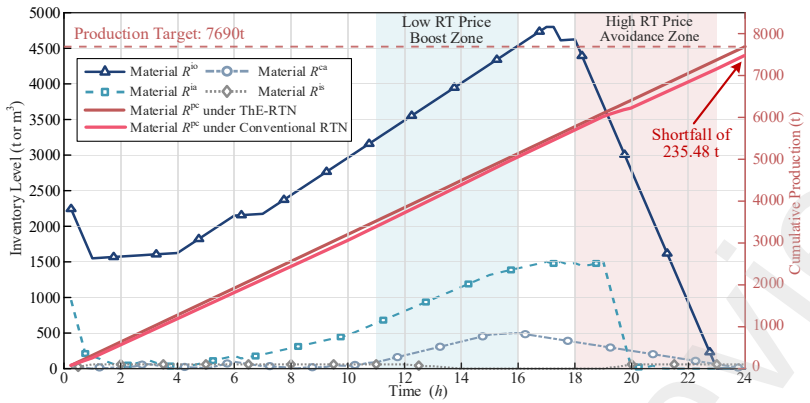


Fig. 10. Material-inventory trajectories under ThE-RTN and cumulative production obtained by thermodynamic replay of the two RTN schedules.

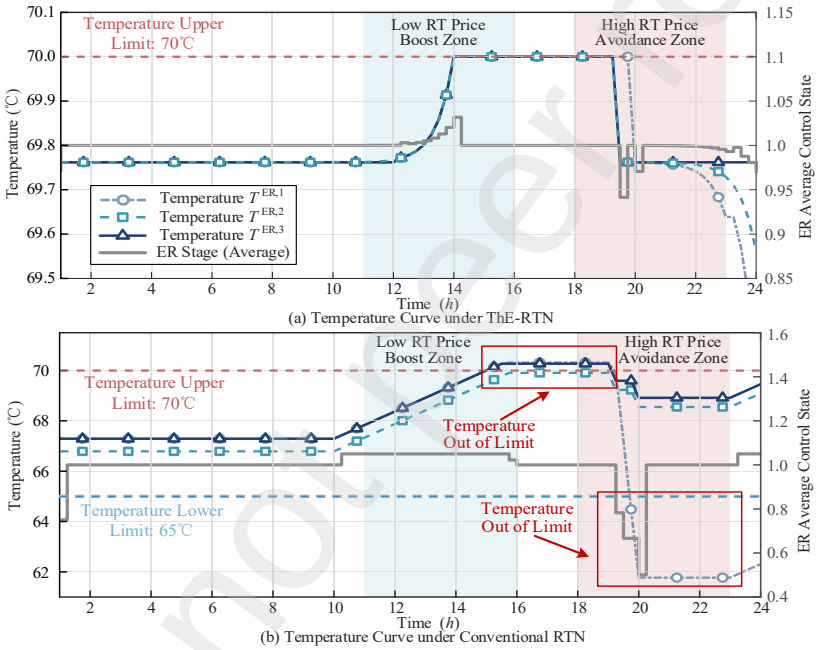


Fig. 11. Electrolyte-temperature trajectories and average ER control states under (a) the ThE-RTN and (b) the conventional RTN.

As shown in Figs. 9 and 10, both models shift load toward low-price periods, but the conventional RTN concentrates deeper curtailments and equipment shutdowns during high-price hours. The ThE-RTN instead coordinates power regulation with material buffers, accumulating inventory before the high-price interval and sustaining downstream production during curtailment. Under thermodynamic replay, the prolonged low-temperature operation induced by the conventional schedule causes cumulative efficiency losses, yielding only 7454.52 t and a production shortfall of 235.48 t.

Fig. 11 explains this production gap. The ThE-RTN maintains the ER temperatures within 65-70 °C, whereas the conventional RTN drives them above the upper limit during low-price boosting and below the lower limit following concentrated curtailment. These violations demonstrate that the conventional RTN overestimates executable flexibility, while the ThE-RTN preserves thermodynamic feasibility and production reliability. Overall, the ThE-RTN converts nominal scheduling flexibility into thermodynamically feasible and production-reliable flexibility.

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Finally, the accuracy of the SOS2-McCormick reformulation is evaluated by comparing the approximation error revealed by nonlinear replay with the theoretical upper bound in Eq. (27). The original SC-IGDT plan yields a terminal production of 7690 t in the MILP model, whereas replaying the same plan using the original nonlinear relationships gives a minimum terminal production of 7675.19 t across all representative PV scenarios. The resulting maximum shortfall is 14.81 t, corresponding to a relative discrepancy of 0.1926%, which is below the theoretical error bound of 0.3954%. Although the replay error remains within the theoretical bound, the residual production shortfall necessitates further compensation to ensure physical executability at the specified production precision.

For nonlinear-replay validation, a terminal-production tolerance of 1 t is predefined, accounting for approximately 0.013% of the 7690 t production target. Accordingly, a nonlinear-replay terminal production of at least 7689 t is regarded as satisfying the production requirement at this precision. For each PV scenario, the difference between the terminal productions obtained from the linear model and nonlinear replay is calculated, and the portion exceeding the prescribed tolerance is added to the corresponding linear terminal-production requirement. The DA bid and production schedule are subsequently reoptimized, with all other model parameters and uncertainty settings unchanged. This replay-and-compensation experiment is conducted independently of the primary economic comparison. Table 7 summarizes the results.

Table 7. Nonlinear replay and terminal-production compensation results.

Metric	SC-IGDT Plan	Compensated Plan
Minimum nonlinear terminal production (t)	7675.19	7689.68
Maximum terminal shortfall (t)	14.810	0.321
Maximum relative shortfall (%)	0.1926	0.0042
Scenarios satisfying the production tolerance	0/6	6/6
Temperature/material/control feasibility	Feasible	Feasible

After re-optimization, the minimum terminal production obtained from nonlinear replay across all scenarios increases to 7689.68 t, while the maximum residual shortfall decreases from 14.81 to 0.321 t, below the prescribed 1-t tolerance. Meanwhile, all temperature, material, and control constraints remain satisfied. The corresponding risk cost increases from 18122.15 CNY to 18206.18 CNY, an increase of only 84.02 CNY, or 0.464%. These results demonstrate that accounting for the linearization-induced production deviation restores production feasibility within the specified tolerance and preserves physical feasibility at a limited additional cost, further confirming the effectiveness of the proposed framework in coordinating economic decisions with physical execution.

5. Conclusion

Under high renewable penetration, grid-interactive PILs must balance economic performance, uncertainty protection, and physical feasibility under PV and price uncertainties. Addressing thermodynamic neglect and decoupled scheduling-risk decisions, we develop a coordinated production-scheduling and bidding framework. ThE-RTN embeds power-temperature-efficiency coupling into plant-wide scheduling, while SC-IGDT manages scenario-based PV and DA-RT price risks under settlement rules. Case studies show that a 10% cost tolerance yields a maximum feasible price-uncertainty radius of 3.781%. SC-IGDT further mitigates extreme costs and reduces budget exceedances under amplified price errors. The proposed framework reduces spot-exposure risk cost by 17.77% relative to conventional RTN scheduling and bidding. Thermodynamic replay verifies production and thermal feasibility, whereas conventional RTN causes thermal violations and a production shortfall. These results demonstrate coordinated improvements in uncertainty protection, economic performance, and operational safety. Future work will investigate intraday rolling optimization and control.

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